

Constructing Repairable Architectures for AI-Assisted PDE Proofs

Local Closure, Residual Promotion, and Routed Obstructions

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May 7, 2026

Abstract

We develop a construction procedure for repairable AI-assisted PDE proofs. The method replaces the search for one decisive global estimate by an iteration of local closure, residual promotion, and certified reduction to previously discharged obstructions. At each state, the proof fixes a local window, scale, representative, quotient data, compactness topology, input contract, and retained obstruction; applies a standard local theorem only to the terms covered by its hypotheses; and promotes every remaining term or failed hypothesis to a named descendant state.

The resulting object is a finite local proof graph whose route edges are local analytic capsules and whose reduction edges certify closure by an already discharged obstruction. The graph records the data needed to audit representative normalizations, compactness passages, imported estimates, terminal closures, and repair after failed estimates. The architectural commitment is a backward-contradiction shape: the routing transports the retained obstruction (the negation of the target theorem) through coordinate changes until an extracted limit profile realizes it, then establishes a terminal contradiction at that single profile rather than by forward propagation of smallness; this architectural choice compresses page count drastically below the forward-tracking idiom. The local Stokes equations provide a closed linear stress test: vorticity and local pressure estimates close the non-harmonic contribution, while the harmonic-gradient remainder becomes a named state closed only after the correct pressure normalization is supplied. A Lean backend checks the finite certificate and the declared analytic boundary; the PDE estimates themselves remain named imported contracts.

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1 Introduction

Critical PDE arguments usually proceed by localizing near a candidate singular region, rescaling, extracting subsequences, fixing representative normalizations or quotient choices, and applying

compactness or rigidity on smaller cylinders. At each step the proof must keep track of the active domain, scale, topology of convergence, normalization of nonlocal variables, hypotheses of imported estimates, and the nonzero quantity that will be contradicted at the terminal step. These data are part of the analysis. A locally coherent generated passage can change the problem being solved by replacing a local hypothesis with a global bound, changing a representative normalization, passing to a limit in a missing convergence topology, or applying a rigidity theorem before its hypotheses have been proved.

When these analytic dependencies are recorded only in prose, AI-assisted drafting tends toward two endpoints: expository summaries of known PDE arguments, or manuscripts whose decisive step is a single global estimate. The second mode is structurally unstable. When that estimate is invalid, stated in the wrong topology, or missing a representative or compactness hypothesis, the error can support many later claims. The manuscript contains no formal containment boundary for the failure.

The mathematical background already contains powerful tools for this task. Concentration compactness records loss of compactness [3, 4]; profile decompositions refine compactness defects into profiles, parameters, and remainders [5, 6, 7]; Littlewood–Paley theory and dyadic decompositions isolate scale and frequency interactions [8, 9]; singular-integral theory supplies local representative and projection estimates used in many PDE arguments [10]. This paper uses these tools as analytic inputs and changes their deployment in an AI-assisted proof: each tool is invoked on the local window and in the topology where its hypotheses have been produced; the component it controls is closed locally; and the profile, tail, defect, representative remainder, bad dyadic packet, or missing topology that remains is promoted to the next state.

The AI and formalization literature supplies another part of the setting. Proof assistants have begun to formalize substantial PDE-adjacent analysis, including HOL Light developments for classical PDE models [11], Lean formalizations of the Gagliardo–Nirenberg–Sobolev inequality [12], tempered distributions and Sobolev spaces [13], higher-order differential calculus [14], and De Giorgi–Nash–Moser theory [15]. In parallel, LLM-assisted theorem proving and autoformalization systems have developed Lean-facing proof search, premise retrieval, and formalization pipelines [16, 17, 18]. PDE-specific LLM systems have mainly targeted control specifications and numerical solver generation [19, 20]. These developments leave a distinct analytic interface: the repairable PDE proof state in which the model operates before all local estimates have been fully formalized.

The contribution of this paper is a method for constructing such arguments. The procedure builds the graph by repeatedly localizing the current state, closing the component controlled by standard estimates, promoting the remaining obstruction to a named state, and selecting a new normalization, quotient, compactness topology, or decomposition from the structure that remains. This local-closure step is the basic model-facing operation: the requested output is a local PDE capsule, a declared hypothesis ledger, and the residual routes left by that capsule.

The resulting graph is a finite local proof graph. Each route edge is a local analytic capsule with a declared local window, scale and topology data, representative or quotient status, input contract, output contract, retained obstruction, failure routes, and audit checks. Each terminal node is marked closed, imported, or **active-frontier** in the development ledger. A finished proof has no reachable **active-frontier** node. A failed estimate is represented as an invalid capsule, a strengthened input contract, a named failure route, or a reduction to a previously discharged obstruction.

The exposition proceeds from construction to audit. It first positions the method as a local deployment discipline for compactness, profile, dyadic, and singular-integral tools, and then as an analytic interface for AI-assisted proof development before full formalization. It next gives the

graph-construction protocol: choose the local coordinate package, close the controlled component, promote the remainder, extract its invariant structure, and choose the next state. The backward-contradiction architecture of [Section 5](#) then makes the framework’s architectural commitment explicit: the routing settles into a backward proof from a retained obstruction through coordinate changes to a terminal contradiction at a single extracted limit profile, rather than by forward propagation of smallness; this architectural choice, contrasted against the dominant forward-tracking idiom, accounts for the observed page-count compression on nonlinear problems. The common PDE failure modes then become adversarial tests for proposed capsules and routes. A failed test returns the affected state to the active frontier, strengthens an input contract, creates a failure route, promotes a residual to a new state, or records a certified reduction to an already closed obstruction.

The paper then defines the local PDE proof object, proves soundness and local invalidation principles, and stress-tests the architecture on local Stokes regularity. This linear closed test already contains a canonical PDE failure mode: vorticity smoothing misses time-dependent harmonic-gradient velocity modes absorbed by pressure. The routed proof records this row and closes the normalized theorem that remains true.

The Lean backend checks the finite proof-state certificate around named analytic imports: state coverage, transition ranking, terminal reachability, pointwise local-to-global assembly, representative and kernel remainder rows, parasitic-row closure, and the explicit analytic boundary. The analytic PDE estimates remain mathematical contracts. The certification claim is that the formal layer prevents an imported local estimate from becoming an undeclared global theorem, while the analytic content is stated and audited as PDE.

2 Mathematical and AI Positioning

The architecture has both a mathematical role and an AI-facing role. On the mathematical side, it assumes the standard PDE toolkit: localization, compactness extraction, concentration compactness, profile decomposition, Littlewood–Paley and dyadic decompositions, Calderon–Zygmund estimates, local regularity criteria, and perturbative closure. Its contribution is a discipline for deploying those tools locally and compositionally. A tool closes the component for which the current state has produced the hypotheses. The remaining component becomes a named state in the graph.

Compactness and decomposition are treated as local construction steps instead of final global closures. Concentration compactness identifies the way compactness can fail; the proof graph records each failure as a local state with a retained obstruction. A profile decomposition separates profiles, parameters, tails, and perturbative remainders; the graph assigns these outputs to different states, so that a compact profile may close locally while a tail, diffuse defect, or missing topology is routed elsewhere. A dyadic decomposition isolates scale or frequency packets; the graph closes the packets controlled by the available estimate and routes the bad packet or scale interaction to a scale state. A Calderon–Zygmund estimate closes the localized singular integral part; harmonic, exterior, or representative-dependent components remain representative data for a later capsule.

The method therefore uses global analytic tools in local form. Many decomposition theorems are stated in a fixed ambient space or for global bounded sequences. In the local proof graph, the theorem is invoked only after a local cylinder, representative, scale, and topology have been selected. The proof records exactly which component closed on that local object and which

component survived. The next state is determined by the mathematical structure of the remainder.

For PDE readers, the object can be read as a local version of the usual compactness ledger. It keeps the concentration, profile, dyadic, representative, and perturbative rows separated until the hypotheses needed to merge or close them have been produced. For AI-assisted work, the same ledger prevents the model from replacing that construction by a single global estimate. For formal checking, it supplies a finite dependency object around the analytic estimates.

Proof assistants and proof-carrying certificates address the checking of specified formal artifacts [1, 2]. Recent Lean work shows that major analytic infrastructure and modern PDE regularity theorems can be machine checked [12, 13, 14, 15]. LLM theorem-proving systems supply proof search, premise retrieval, and autoformalization support inside formal environments [16, 17, 18]. PDE-specific LLM systems demonstrate that language models can help translate, reason about, and program PDE control or solver tasks [19, 20].

This paper addresses the analytic layer between these strands. The object is a developing PDE argument before full formalization: local regimes, compactness passages, representative choices, imported estimates, retained obstructions, and failure alternatives. A generated estimate can change the analytic problem by replacing local hypotheses with a global bound, changing a representative normalization, passing to a limit in a topology that has not been obtained, or invoking a terminal theorem before the relevant obstruction has been carried to the limiting problem.

The resulting artifact is a finite dependency graph of local PDE claims whose estimates, normalizations, topologies, obstructions, imports, reduction edges, and unresolved alternatives are available for audit and repair.

3 Design Requirements for AI-Assisted PDE Proof Construction

PDE estimates are applied inside a structured analytic state. A local argument fixes a space-time window, a scale, cutoff functions, representative normalizations, quotient choices, compactness topology, imported hypotheses, and the obstruction that must be carried to the terminal contradiction. In a conventional manuscript, much of this structure is maintained by the author through notation and context. In AI-assisted drafting, the same structure must be recorded as part of the proof state, because a locally plausible estimate can change the analytic problem being solved.

The basic object of the architecture is therefore a local construction state. Such a state records the cylinder or ball on which the next estimate is to be applied, the scale at which the estimate is normalized, the representative chosen for nonlocal or kernel quantities, the quotient or kernel modulo which compactness is taken, the topology available for passage to limits, and the retained obstruction that must survive the local step.

The proof graph must satisfy five design requirements.

- (R1) **Coordinate-first matching.** Every local theorem is matched against a recorded coordinate package rather than against prose.
- (R2) **Declared input contracts.** The input contract for a theorem is recorded before the theorem is applied.
- (R3) **Residual promotion.** Every uncovered representative term, tail term, commutator, compactness defect, or missing convergence statement is promoted to a descendant state with a stable signature.

(R4) Explicit closure witnesses. A descendant closes only by a local capsule, by an imported theorem, or by a certified reduction edge to a previously discharged obstruction.

(R5) Frontier accounting. A partial development distinguishes active frontier states from frozen open ledger obligations.

These requirements make generation a constrained theorem-matching task. The model retrieves and states a local PDE estimate, applies it only to the terms covered by its hypotheses, and identifies the residual terms or failed hypotheses that remain outside the theorem. The resulting repair mechanism is local: a failed estimate becomes a named residual state with a bounded dependency cone and a specified analytic signature. The manuscript, automated evaluator, and Lean certificate can audit whether each local theorem is applied under its stated hypotheses, whether residual terms are transported rather than discarded, and whether terminal closure uses exactly the obstructions and topologies produced along the route.

4 Constructing the Local Proof Graph

The local proof graph is produced by an iterative local-closure protocol. A state is a local PDE regime together with the obstruction that must survive through the next step. The construction step has three parts. First, choose the local coordinates in which a standard estimate can be invoked. Second, close only the component controlled by that estimate on those coordinates. Third, promote the remaining structure to the next state.

4.1 Choosing the Local State Coordinates

Coordinate selection is part of the proof. The current branch supplies a retained obstruction, local bounds, available representatives, and upstream failure alternatives. The coordinate step records the smallest local setting in which one standard PDE estimate, compactness theorem, decomposition, or regularity criterion can be applied without changing the problem.

Definition 4.1 (Coordinate package). A coordinate package for a local state S and a proposed capsule is

$$\Gamma(S) = (z_0, r, W_{\text{core}} \subseteq W_{\text{work}}, \pi, \mathcal{Q}, \tau, I, H).$$

Here z_0 is the selected spacetime center, r is the selected scale, $W_{\text{core}} \subseteq W_{\text{work}}$ are nested local windows, π is the representative for nonlocal or kernel quantities, $\mathcal{Q}$ records quotient or kernel data, τ is the compactness topology, I is the retained obstruction, and H is the input contract available at the state. The package is admissible when every component is produced by current branch data or by a named upstream capsule, and every missing component has a named failure route.

The choice of $\Gamma(S)$ is forced by the obstruction and by the first local theorem to be used. The center and scale are selected where the obstruction is visible in a scale-invariant quantity. The working window is large enough to support cutoffs, representative estimates, compactness, and a conclusion on the core window. The representative, quotient data, and compactness topology are the ones required by the local theorem and produced by the available input contract. When any coordinate is unavailable, the branch stays on the active frontier and routes to the corresponding failure state. [Table 1](#) gives the construction rule.

Table 1: State-coordinate selection for a local capsule.

Coordinate	Selection rule	Audit check and failure route
Retained obstruction I	Record the nonzero, nonsmooth, noncompact, or otherwise terminally incompatible quantity supplied by the negated theorem or by an upstream capsule.	The obstruction must have a local lower bound or an invariant formulation. If it vanishes in the selected window, route to the appropriate local regularity, vanishing, or no-obstruction closure.
Center and local cylinder	Center the core window at a point or packet where the retained obstruction is visible; choose a working cylinder that contains the core with buffer for cutoffs and smaller-cylinder conclusions.	Every estimate must consume data on the working window and conclude on the core. If the obstruction leaves every compact core, route to recentering, exterior escape, diffuse defect, or profile selection.
Scale r	Select the scale at which a critical or invariant quantity is normalized, first becomes active, or retains a fixed positive amount.	Check that all capsule terms have the declared critical dimension. If no stable scale is selected, route to scale instability, multiscale interaction, cascade, or exterior state.
Nonlocal representative π	Use the representative required by the local equation: for example a mean-subtracted field, a harmonic/local split, or a representative modulo functions invisible to the equation.	The representative must be compatible with restriction, extraction, and the downstream estimate. If the representative row is unavailable, route to a representative-normalization, kernel-remainder, or compactness state.
Quotient or kernel data $\mathcal{Q}$	Identify constants, harmonic components, affine modes, local kernels, or other representative freedoms before applying a recovery theorem.	The output is quotient-level unless a later compatibility capsule recovers the raw representative. If the kernel component is not controlled, route to a quotient state or normalized-representative state.
Compactness topology τ	Choose a topology produced by the available local bounds and strong enough for the next nonlinear passage, representative passage, or obstruction-retention step.	Every limit passage must use this topology explicitly. If the available topology is too weak for the next operation, route to a compactness-defect, diffuse-loss, exterior-escape, or missing-topology state.

Table 1: State-coordinate selection for a local capsule (continued).

Coordinate	Selection rule	Audit check and failure route
First local theorem	Choose the local estimate, compactness theorem, decomposition, or regularity criterion whose hypotheses are contained in the input contract H .	The capsule may close only the component covered by that theorem. Any missing hypothesis becomes a strengthened input contract, a failure route, or a descendant state on the active frontier.

Note. This table implements the coordinate-selection step of the construction algorithm. Its output is an admissible local coordinate package $\Gamma(S)$. Only after this package has been supplied may the proof invoke a local capsule and then apply the residual-promotion rule in [Table 2](#).

4.2 Local Closure

Local closure is the step that turns the coordinate package into a checked mathematical move. Once $\Gamma(S)$ has been fixed, the proof searches the local literature for one estimate, compactness theorem, decomposition, or regularity criterion whose hypotheses match the declared input contract H . The theorem is then applied only on the working window and only to the terms covered by those hypotheses. Its conclusion is recorded as the capsule output O , with the same representative, quotient data, topology, and retained obstruction declared in $\Gamma(S)$.

The closure rule keeps the scope of each local theorem tied to its hypotheses. A standard energy or smoothing estimate closes the principal component on a smaller cylinder; a localized representative estimate closes the representative component; a compactness theorem produces a limit only in the topology stated in the input contract; a div-curl theorem recovers only the quotient-level representative allowed by its kernel. Terms outside the stated hypotheses become residual data for the next construction step.

4.3 Residual Promotion

After the coordinate package has been fixed, the essential operation is residual promotion. Suppose a state has local data

$$(W, \Sigma, H, I),$$

where W is the spacetime window, Σ records scale, topology, representative or quotient data, H is the available hypothesis contract, and I is the retained obstruction. A proposed estimate or compactness passage decomposes the state into a controlled component and a residual component,

$$\text{state data} = \text{controlled local contribution} + \text{residual contribution}.$$

The controlled contribution may be closed only by a named estimate whose hypotheses are contained in H . The residual contribution becomes a descendant state with its own window, topology, representative data, retained obstruction, and failure routes.

The constructive step is to read structure from the residual. A promoted residual is assigned a stable signature

$$\text{sig}(R) = (\text{scale, locality, representative freedom,} \\ \text{compactness topology, retained obstruction, defect type}).$$

The next normalization, quotient, topology, or decomposition is chosen to make one component of this signature explicit. The graph records successive local closures and the structures that survive those closures. Table 2 summarizes the corresponding construction choices.

Table 2: Residual signatures used to construct descendant states.

What remains after local closure	Stable structure to record	Next state to construct
Nonlocal representative or tail term	Local part, kernel remainder, exterior contribution, or quotient ambiguity	Representative state, exterior state, or quotient-recovery capsule.
Failure of a compactness passage	Missing topology, diffuse loss, exterior escape, or weak limit insufficient for the next estimate	Compactness-defect state with the exact topology required by the downstream closure.
Ambiguity of representative	Invariant quotient, removable local mode, or component seen only after normalization	Normalized-representative state or quotient state, with a compatibility route back to raw variables.
Instability of center or scale	Selected center, selected scale, modulation parameter, or separated scale family	Recentered state, rescaled state, multiscale state, or scale-collapse state.
Several active local packets	Compact core, separated profiles, comparable-scale packets, or same-point cascade	Profile-decomposition state, multibubble state, cascade state, or componentwise local closure.
Bad dyadic packet or scale interaction	Localized frequency scale, high-low interaction, paraproduct error, or commutator defect	Dyadic state, commutator capsule, scale-interaction state, or perturbative remainder state.
Mass or energy outside fixed compact windows	Exterior escape, diffuse measure, annular concentration, or tail recurrence	Exterior state, diffuse-defect state, annular-tail state, or terminal residual state.
Monotonicity or energy identity leaves uncompensated terms	Canonical cost, transition term, scale-drift term, or shell-work term	Cost-validation state and localized-error states for the nonintegrable components.
Terminal residual with no lower local class available	Realized terminal profile, perturbative remainder, or unresolved obstruction	Terminal-profile state or perturbative closure capsule; absent such a capsule, the route remains a partial-proof obligation.

Note. Use this table as a construction rule. After a local estimate closes the controlled contribution, the first column identifies the unclosed residual, the second column records the structure retained by that residual, and the third column names the next state for which a local capsule must be supplied.

Algorithm 1 summarizes the constructive core of the paper. It replaces the search for a single global estimate by repeated local theorem matching, residual promotion, and adversarial audit.

Algorithm 1. Local closure and residual promotion

Input: Target theorem T , branch $B_0 \models \neg T$, retained obstruction I_0 , and local theorem library $\mathcal{L}$.
Output: Finite local proof graph G , with terminal statuses **closed**, **imported**, or partial-ledger **active-frontier**.

- 1 Initialize G with the root state determined by B_0 and I_0 ; put the root state on the active frontier
- 2 **while** *the active frontier contains a state S* **do**
- 3 Choose the coordinate package $\Gamma(S)$ using Table 1
- 4 Search $\mathcal{L}$ for a local estimate, compactness theorem, decomposition, or regularity criterion matching $\Gamma(S)$
- 5 Record the theorem hypotheses as the capsule input contract
- 6 Apply the theorem only to the terms covered by that input contract
- 7 Promote every uncovered term, failed hypothesis, or missing topology to named descendants $S'_1, \dots, S'_m$
- 8 **foreach** descendant S'_j **do**
- 9 Assign its stable signature: scale, locality, representative freedom, compactness topology, retained obstruction, and defect type
- 10 **if** *a matching local closure capsule or imported theorem has been supplied* **then**
- 11 Mark S'_j **closed** or **imported**
- 12 **else**
- 13 **if** *there is a previously closed or imported state S_k such that $I(S'_j)$ reduces to $I(S_k)$ through a valid acyclic reduction edge* **then**
- 14 Add the reduction edge $S'_j \rightarrow_{\text{red}} S_k$ and mark S'_j **closed** by reduction
- 15 **else**
- 16 Choose the next normalization, quotient, topology, or decomposition using Table 2
- 17 Return S'_j to the active frontier
- 18 **Return** G . *A node carries **active-frontier** only in a partial development ledger, when no closure, import, reduction edge, or further stratification has yet been supplied.*

4.4 How the Construction Steps Are Carried Out

Step 1 fixes the proof around a transported obstruction. The obstruction is the quantity that the terminal closure must contradict: retained concentration, retained mass, failed regularity, noncompactness, or an incompatible terminal profile. Recording it first prevents the construction from being organized by the desired conclusion alone.

Step 2 turns the branch into a local matching problem. The coordinate package $\Gamma(S)$ selects the cylinder, scale, representative, quotient, topology, input contract, and obstruction for the current state. These choices are taken from branch data and named upstream capsules. If one coordinate is not available, the branch is routed to the corresponding failure state before any local theorem is invoked.

Step 3 is the literature-retrieval step. Once the coordinate package is fixed, the model searches the existing PDE toolkit for a theorem whose domain, scaling, representative, topology, and boundary status match the current state. This task matches current strengths of language models: retrieving candidate estimates, comparing hypotheses, drafting local theorem statements, and proposing nearby standard decompositions. The expected output is a candidate local theorem together with its hypothesis ledger.

Step 4 enforces the hypothesis ledger. The chosen theorem is applied only to the terms covered by its stated assumptions. A local estimate may close the interior contribution, a representative estimate may close the declared representative, or a compactness theorem may pass to a limit in the declared topology. Terms outside those hypotheses remain in the proof state.

Steps 5–7 convert failure into construction data. Every uncovered term, failed hypothesis, or missing topology becomes a descendant state with a stable signature. The signature records the structure that survived local closure: scale, locality, representative freedom, compactness topology, retained obstruction, and defect type. The next normalization, quotient, topology, or decomposition is selected from that signature.

Steps 8–9 separate closure from development. A route is closed when a local closure capsule or imported theorem has matching inputs, or when it reduces to a previously discharged obstruction. Such a reduction is a certified internal edge $S' \rightarrow S_k$: the target state S_k is already **closed** or **imported**, the obstruction of S' transports to the obstruction of S_k , the scale, locality, representative, quotient data, and compactness topology are compatible, and the edge decreases the routing rank or is otherwise acyclic. If no closure, import, reduction, or further stratification is available, the descendant remains on the active frontier. An **active-frontier** node records an unfinished obligation in a partial ledger and carries no closure status in a completed proof.

The method constrains generation to declared PDE operations: local estimates on declared cylinders, standard compactness passages, known decompositions, and precise bookkeeping of hypotheses. When the standard machinery closes the local contribution, the graph records the closure. When it fails, the failure becomes mathematical data. The next state is selected from the structure of that data rather than from an invented global bound.

4.5 Operational Protocol

The operational protocol is the execution form of [Algorithm 1](#), with the adversarial checks of [Section 6](#) applied after each proposed capsule. The proof begins with a target obstruction. At each step the current state declares its local window, scale, topology, representative or quotient status, input contract, output contract, and retained obstruction. The model is then asked for one local capsule: an estimate, compactness passage, normalization, decomposition, or closure theorem whose hypotheses are visible in the current state.

The capsule is accepted only after its local contract has been checked. Each compactness passage records the topology in which the obstruction survives. Each representative step records the local representative and kernel remainder. Each terminal theorem is invoked after its local hypothesis ledger has been produced. Each imported theorem carries explicit boundary status. A route is closed when it reaches a closure capsule, an imported theorem with matching hypotheses, or a valid acyclic reduction to a previously discharged obstruction.

Reduction is a separate closure witness. When a residual state matches a configuration already excluded earlier in the graph, the proof records a reduction edge rather than treating the name match as closure. The edge states the target closed or imported node, the map transporting the retained obstruction, the compatibility of local windows, scales, representative representatives, quotient data, and compactness topology, and the acyclicity or strict-rank witness.

Repair uses the same structure. When a capsule fails, the proof invalidates that capsule, recomputes its dependency cone, and preserves closed paths that avoid the cone. The failed capsule is then replaced, strengthened, split into more precise states, or connected by a valid reduction edge. If no closure or reduction is available, the affected state returns to the active frontier; if development stops there, the partial ledger records an open obligation.

5 The Backward-Contradiction Architecture

The construction protocol of [Section 4](#) is shape-neutral: it routes a retained obstruction through coordinate changes, theorem applications, and residual promotions, terminating when every leaf is closed by an import, a direct estimate, a reduction, or a named obstruction. The shape-neutrality is intentional — the same protocol should apply to existence proofs, regularity proofs, blowup-exclusion proofs, and rigidity statements. But on nonlinear PDE problems, the protocol’s output differs sharply depending on which architectural shape the routing settles into. This section describes the architectural commitment that the framework makes, why it differs from the dominant proof shape in the modern PDE literature, and how that difference produces the page-count compression observed in the worked examples.

5.1 The proof shape

The framework’s proof architecture is *backward contradiction at an extracted profile*:

1. The root retained obstruction I_0 is the **negation** of the target theorem — the assumed pathology, blowup, singularity, loss of compactness, or failure of the desired conclusion.
2. The routing transports I_0 through coordinate changes (rescaling, gauge fixing, frequency localization, frame change) until a state is reached at which the obstruction is *realized at a single limit profile* Φ_* , extracted by compactness (Banach-Alaoglu, time-translation, concentration, profile decomposition).
3. The reduced equation satisfied by Φ_* is derived by passing the original equation to the limit, recording obstruction transport at every coordinate change.
4. The terminal contradiction is a textbook estimate evaluated **at the single profile** Φ_* , contradicting the threshold value carried by the retained obstruction.

This is not the same as the dominant shape of nonlinear PDE proofs in the modern literature, which is *forward propagation of smallness*: define a quantity $\mathcal{Q}[u(t)]$ that is small initially, prove a differential or integral inequality that controls $\mathcal{Q}[u(t)]$ for all t in some interval, conclude that the solution remains in the regime where the inequality holds (continuity / bootstrap argument), and deduce the target theorem as a corollary. Forward propagation requires the nonlinear coupling to be controlled as an *operator* on a function space — typically through paraproduct estimates, Newton or Nash-Moser iteration in scales of regularity, or contraction of a fixed-point map. The operator-level control is what produces the page count: bounding a bilinear cascade as an operator over all admissible inputs is fundamentally more work than bounding it at one point.

The backward-contradiction architecture eliminates the operator-level work. The bilinear, nonlinear, or critical residual term is evaluated at the *one* profile Φ_* rather than as an operator. The terminal estimate has the form

$$\|\Phi_*\|_\tau \leq C\varepsilon + C'\|\Phi_*\|_\tau^k, \quad k \geq 2,$$

and the contradiction arises from elementary algebra of the corresponding scalar inequality: the threshold value of $\|\Phi_*\|_\tau$ falls in the interval forbidden by the parabola, so the assumed pathology cannot occur. There is no cascade to sum, no operator norm to track, no fixed-point iteration, and no time interval to control.

5.2 The three terminal contradiction types

Every closed terminal in a framework-compliant proof is one of three contradiction types. Forward-tracking proofs use a fourth type (Grönwall-type continuation) that the framework forbids at terminals.

- (T1) **Quadratic self-bound at the profile.** The reduced equation gives $\|\Phi_*\|_\tau \leq C\varepsilon + C'\|\Phi_*\|_\tau^k$ for some $k \geq 2$. The associated parabola $C'u^k - u + C\varepsilon$ is nonnegative outside an interval $[u_-, u_+]$. The threshold value of $\|\Phi_*\|_\tau$, transported from the routing, falls in the forbidden interval (u_-, u_+) for sufficiently small ε . Contradiction.
- (T2) **Conservation comparison.** A conserved invariant $\mathcal{H}[f]$ has an upper bound $\mathcal{H}[f(t)] \leq C\varepsilon^k$ from initial smallness and a lower bound $\mathcal{H}[\Phi_*] \geq c \cdot |\Phi_*|^k$ from the geometry of Φ_* (rearrangement inequality, isoperimetric estimate, level-set area). The two bounds contradict at the extracted profile.
- (T3) **Rigidity at the extracted profile.** The extracted profile is bounded ancient, self-similar, or otherwise globally constrained, and a textbook rigidity theorem (Liouville, classification of ground states, classification of self-similar profiles) forces $\Phi_* = 0$. The retained obstruction asserts $\Phi_* \neq 0$, contradicting rigidity.

The forbidden type — the one the framework excludes — is **continuation by Grönwall**: showing that a quantity stays small over a time interval by a differential inequality $\frac{d}{dt} \mathcal{Q} \leq F(\mathcal{Q})$ closed via $\mathcal{Q}(0)$ small. Continuation is the engine of forward-tracking proofs; it is not a terminal contradiction at a profile.

5.3 Algorithm 1' with the extraction step

The construction algorithm of [Section 4](#) is correct but does not make the extraction step explicit. We rewrite it here with the extraction step inserted, since it is the architectural keystone.

Algorithm 2. Local closure with profile extraction

Input: Target theorem T ; root branch B_0 with retained obstruction I_0 defined as the negation of T 's conclusion; library $\mathcal{L}$ of textbook theorems with named citations.

Output: Finite directed acyclic graph (DAG) G with terminals labeled by closure type.

- 1 Initialize root S_0 with obstruction I_0 ; set active frontier $F = \{S_0\}$
- 2 **while** F is nonempty **do**
- 3 Select state S of lowest rank from F
- 4 Fill coordinate package $\Gamma(S)$ using Table 1
- 5 Search $\mathcal{L}$ for a single local theorem matching $\Gamma(S)$ and the input contract $H(S)$
- 6 **if** a matching theorem exists **then**
- 7 Apply it to covered terms only; promote each residual to a descendant via Table 2; record the explicit obstruction-transport functional for each route edge
- 8 **else**
- 9 **if** the residual is a bilinear, nonlinear, or critical coupling not closed uniformly by a single textbook theorem **then**
- 10 (Profile extraction) Identify the appropriate compactness mechanism (Banach-Alaoglu, time-translation, concentration, profile decomposition) and extract a single limit profile Φ_* realizing the obstruction. Audit obstruction transport: provide an explicit dual functional continuous in the weak-* topology. Derive the reduced equation at Φ_* and route each component (source, linear, nonlinear) as separate estimates at the single profile
- 11 **else**
- 12 **if** no Table 2 row matches the residual signature **then**
- 13 Terminate the route as closed-by-named-obstruction
- 14 **else**
- 15 Return S to the active frontier for further decomposition
- 16 Verify acyclicity of the resulting DAG; verify that every terminal contradiction is of type (T1), (T2), or (T3) (Section 5.2). Reject any terminal closure of Grönwall-continuation type
- 17 **Return** G .

The extraction step (step 2.5 in the pseudo-code above) is what distinguishes Algorithm 1' from a generic theorem-routing protocol. Without it, the natural completion of step 2.4 on a nonlinear residual is to invoke an operator estimate — paraproduct, multilinear interpolation, fixed-point setup — which immediately reintroduces the cascade machinery that the architecture is designed to avoid. With it, the algorithm forces the routing into the backward-contradiction shape.

5.4 Diagnostic flags for forward-tracking creep

The forward-tracking shape is the dominant idiom in modern nonlinear PDE. A practitioner — human or model — running the framework will revert to forward tracking by default unless explicit smell tests are applied. We list the diagnostic flags. If a proof under construction exhibits any of the following, the routing has drifted into forward tracking and must be rerouted via Section 5.3 step 2.5.

1. **Time-interval supremum.** A statement of the form “show $\sup_{0 \leq t \leq T} \|u(t)\|_X \leq C\varepsilon$.” The framework controls one profile, not an interval.
2. **Contraction map.** Defining an operator $\Phi : B \rightarrow B$ on a Banach ball and showing it contracts. The bilinear is then iterated; the proof works by Banach fixed point. The framework evaluates the bilinear at one point.

3. **Operator-norm bilinear estimate.** A statement of the form $\|\mathcal{B}\|_{Y \otimes Y \rightarrow Y} \leq C$ where $\mathcal{B}$ is a bilinear map. The framework needs $\|\mathcal{B}[\Phi_*, \Phi_*]\|_\tau \leq C\|\Phi_*\|_\tau^2$ at the single profile, which is strictly weaker.
4. **Paraproduct decomposition as an operator.** Bony's paraproduct produces a continuous bilinear operator on Besov spaces. Used at one profile, the paraproduct is a kernel decomposition; used across all admissible inputs, it is a cascade. The framework requires the former.
5. **Newton or Nash-Moser iteration.** Schemes that linearize the equation around an approximate solution and iterate, recovering regularity loss at each step via tame estimates. These are the canonical examples of forward tracking in scales of regularity.
6. **Shrinking or time-dependent norms.** A time-dependent norm or analytic radius $\lambda(t)$ chosen to absorb cumulative regularity loss. The framework's routing has fixed norms at the extracted profile; time-dependent norms are a forward-tracking artifact.
7. **Bootstrap or continuity argument.** A scheme showing that the set $\{T : \|u(t)\|_X \leq M\varepsilon \text{ for all } t \leq T\}$ is open, closed, and nonempty. This is the canonical shape of forward continuation.

A useful operational rule: if the proof's terminal contradiction is at a *time* (final T , blowup time T^* , or the limit $T \rightarrow \infty$) rather than at a *profile* (a single Fourier configuration Φ_* extracted from a sequence), the architecture is forward tracking.

5.5 Operator-level versus single-profile estimates

The architectural difference between forward tracking and backward contradiction lies in a fundamental choice: whether the nonlinear terms are estimated as operators on function spaces or as evaluations at a single extracted limit profile.

Forward-tracking architectures require operator-level control. The nonlinear bilinear coupling, for instance, must satisfy an estimate of the form

$$\|\mathcal{B}\|_{Y \otimes Y \rightarrow Y} \leq C,$$

which bounds the operator uniformly over all admissible input pairs. This operator-level control is necessary when the proof propagates smallness forward in time: at every intermediate stage, the solution's norm must be bounded over all times in the current interval, so every nonlinear term must be controlled as an operator on that time-slice data.

Backward-contradiction architectures, by contrast, evaluate nonlinear terms only at the extracted limit profile Φ_* . The required estimate is

$$\|\mathcal{B}[\Phi_*, \Phi_*]\|_\tau \leq C\|\Phi_*\|_\tau^2,$$

which is strictly weaker. No cascade machinery is required; the bilinear is a pointwise scalar evaluation, not an operator norm. The terminal contradiction arrives from elementary algebra: the threshold value of $\|\Phi_*\|_\tau$ falls in the interval forbidden by the parabola $Cu^k - u + C\varepsilon$.

This reduction from operator estimates to single-profile estimates is the source of the page-count compression. Operator-level control requires bounding every term in the cascade at every

intermediate scale and interacting all terms under the covering of the entire time axis. Single-profile estimates need only bound the terms at one point. For problems where both architectures are mathematically valid, the backward-contradiction shape produces proofs orders of magnitude shorter.

The [Section 9](#) linear stress test demonstrates the routing protocol on a linear problem. On the Stokes system, the nonlinear terms are absent, so the distinction between operator and single-profile control does not arise. Nonlinear problems (such as reaction-diffusion, waves, or fluid dynamics) are where the architectural choice becomes decisive and the page-count gap becomes visible.

5.6 What the framework adds

The reader may object that profile extraction is a standard technique in nonlinear PDE — Lions’s concentration-compactness, Kenig-Merle’s profile decomposition, Bahouri-Gérard’s wave-equation profile theory — and that the framework merely repackages an existing method. We make three responses.

First, the existing techniques apply profile extraction *within* a forward-tracking architecture. A typical Kenig-Merle argument proves a forward bound (energy nonincrease, scattering criterion) and then uses concentration-compactness to extract a critical profile that violates the bound. The terminal contradiction is at a profile, but the *proof’s spine* is a forward time evolution. The framework’s claim is that on a class of nonlinear problems, the entire proof can be routed *backward from* the extracted profile, with no forward time evolution at all. The forward steps in those problems are artifacts of the dominant idiom, not of the mathematics.

Second, the framework provides the routing language that makes profile extraction available as a default at every nonlinear residual, not as a specialized technique reserved for compactness-defect problems. The [Table 2](#) row “bilinear or quadratic residual at a single extracted limit profile” applies to any state where the routing produces a nonlinear self-coupling. In conjunction with [Section 5.3](#) step 2.5, this turns profile extraction from an occasional technique into the standard move whenever no single textbook theorem closes a nonlinear residual operator-uniformly.

Third, the framework’s explicit prohibition on Grönwall-continuation terminals ([Section 5.2](#)) and the diagnostic flag list ([Section 5.4](#)) prevent the routing from collapsing back into forward tracking at the last step. Without these guardrails, even a profile-extraction-based proof tends to end with “now propagate the bound forward to all time,” reintroducing the cascade machinery the architecture was meant to eliminate.

The framework’s contribution to the analytical content is zero; every leaf is a textbook citation. The framework’s contribution to the architectural shape is to constrain that content into a backward-contradiction graph rooted at the assumed pathology, with profile extraction as the structural bridge between coordinate routing and terminal contradiction. The page-count compression achieved by single-profile estimates versus operator-level cascade control is the architectural payoff. Whether such compressions occur on other nonlinear PDE problems is the empirical question the framework invites.

6 Adversarial Evaluation: PDE Failure Modes and Routed Repairs

Adversarial evaluation is part of construction. Each proposed capsule is tested against the failure modes that commonly break generated PDE arguments before it receives a closure status. The evaluator checks whether the capsule has changed the local hypotheses, changed the representative, used a compactness topology that the state has not produced, invoked a closure theorem before its input contract is available, or absorbed a residual term without a named estimate.

The test decides the next graph operation. A passing capsule closes the current contribution, imports a theorem with matching hypotheses, or adds a valid reduction edge to a previously discharged obstruction. A failing capsule returns construction data: the input contract is strengthened, the residual is promoted to a descendant state, a missing topology or representative is added to the active frontier, or a named failure route is inserted. The graph is then rebuilt locally and tested again. The process stops only when every reachable route is closed, imported, or reduced to a previously discharged obstruction; otherwise the remaining terminal is recorded as an open obligation in the partial ledger.

The loop can be run as a manuscript-level automated evaluator. It checks the structure around an imported PDE estimate: declared inputs, output contract, topology, representative, retained obstruction, residual accounting, and downstream dependency cone. The analytic proof of the imported estimate remains a named mathematical contract. [Table 3](#) lists the adversarial tests used in this paper.

1. Propose one local capsule for the active state.
2. Check that every global or exterior input appears as a declared hypothesis.
3. Check that the representative, quotient, and normalization data are unchanged or explicitly transported.
4. Check that each compactness passage declares the topology in which the retained obstruction survives.
5. Check that each terminal theorem is invoked only after its input contract has been produced.
6. Check that each residual term is either dominated by a named estimate or promoted to a descendant state with a stable signature.
7. Check that any reduction to a previous obstruction records the target state, obstruction transport, compatibility data, and acyclicity witness.
8. Repair locally by strengthening the contract, adding a failure route, splitting the state, importing a theorem, adding a valid reduction edge, or returning the unresolved descendant to the active frontier.

Table 3: PDE adversarial tests and local repair mechanisms.

Adversarial test	Failure detected in an unrouted generated draft	Local repair action
Route name counted as closure	A case is named elsewhere and then treated as solved, or the certificate uses <code>closure_status: route</code> , which is not a valid closure status.	The outgoing edge must be classified as either <code>construction-route</code> (creating a new child capsule, status <code>active-frontier</code>) or <code>obstruction-reduction</code> (transporting to a previously discharged obstruction, status <code>closed-by-reduction</code>). A router node whose every outgoing branch terminates by terminal contradiction or reduction receives status <code>closed-by-branch-exhaustion</code> with a complete branch table; otherwise the target state stays on the active frontier with status <code>active-frontier</code> .
Invented global estimate	Local hypotheses are replaced by global control without a matching input contract.	Global inputs enter only as declared hypotheses; otherwise the local case routes to local, exterior, or compactness-defect states.
Regularity criterion weakened to boundedness	Controlled quantity is used without the matching coupled quantity or scale-critical data.	The regularity capsule states the exact local criterion; missing coupled quantity or critical data route to a representative or scale-defect state.
Representative normalization drift	Mean subtraction, kernel tails, or quotient choices change across limits.	The representative capsule records the local representative, quotient, kernel remainder, and failure route.
Local-to-global overreach	The available topology is compact-window, normalized, or quotient-level.	The target theorem must match the available domain and topology; otherwise the local case routes to a compactness-defect, exterior-escape, or profile state.
Retained mass lost under extraction	The terminal contradiction uses an obstruction no longer proved to survive.	Every extraction capsule declares the transported obstruction.
Escape to infinity omitted	Vanishing on fixed compact sets is promoted to global vanishing.	Compactness capsules include exterior and diffuse escape alternatives.

Table 3: PDE adversarial tests and local repair mechanisms (continued).

Adversarial test	Failure detected in an unrouted generated draft	Local repair action
Rigidity invoked too early	The closure theorem is called before its hypotheses have been produced.	The interface ledger checks source outputs against target inputs.
Circular estimate order	Upper compactness estimates and lower nontriviality floors justify each other.	The graph carries a rank or dependency order separating estimate production from obstruction retention.
Residual terms absorbed without a theorem	A residual term is declared lower order without a named bound.	Each residual term is dominated by a named capsule or promoted to the active frontier with a stable signature.

Note. Each row is an adversarial check on a proposed local capsule. The second column identifies the analytic dependency that the generated text has lost; the third column records the local repair action that keeps the defect inside the local proof graph.

7 The Local PDE Proof Object

After the construction protocol has promoted the residual alternatives, the proof state has the following formal shape. The definitions below record what must be present for each local step to be auditable and repairable.

Definition 7.1 (Branch). A branch at a state is a candidate PDE counterexample trajectory together with the local analytic data retained after all previous route-construction steps and the obstruction assigned to that state.

Definition 7.2 (State). A state is a named local PDE regime characterized by a spacetime window, scale, observables, compactness topology, representative or quotient data, and declared failure alternatives. A state is admissible for route construction when it is stable under the rescalings, subsequences, quotient choices, and normalizations used before its closure theorem is invoked.

Definition 7.3 (Active frontier and partial ledger). The active frontier of a developing proof graph is the set of unresolved states whose input contracts have been produced but whose local closure, obstruction reduction, residual promotion, or descendant route step has not yet been discharged. When development is frozen into a partial ledger, unresolved frontier states are recorded as terminals with status **active-frontier**.

States are local objects. Their definitions may include compact windows, normalizations, or quotient representatives. Their admissible inputs are the declared branch data. Exterior boundary data or whole-space estimates enter only through named capsule hypotheses. This locality condition is the first barrier against global-estimate collapse.

Definition 7.4 (Retained obstruction). A retained obstruction is a nonzero, nonsmooth, or otherwise terminally incompatible quantity attached to a branch and required to survive every allowed extraction, rescaling, normalization change, quotient operation, or descendant construction.

The retained obstruction is the object contradicted at the terminal node. It also makes compactness arguments auditable. A terminal theorem for an auxiliary object closes the original branch only when the proof records that the auxiliary object is realized by the branch and carries the transported obstruction.

Definition 7.5 (Local analytic capsule). A local analytic capsule is a local PDE contract

$$\mathfrak{C} = (W, \Sigma, H, O, I, F, A),$$

where W is the spacetime window, Σ records scale, topology, representative status and quotient data, H is the input contract, O is the output contract, I is the retained obstruction statement, F is the list of named failure routes, and A is the audit checklist.

A capsule may prove an estimate, perform a compactness extraction, construct a normalization, route a failed hypothesis, prove a reduction, or close a terminal state. Its contract specifies the local data it consumes: domain, scale, topology, normalization, obstruction, and imported estimates. An undeclared global hypothesis is a contract failure. A failed capsule remains part of the local PDE proof object: it becomes an invalid edge, an unresolved frontier state, or the source of a new state.

Definition 7.6 (Obstruction-reduction edge). An obstruction-reduction edge from a state S to a previously discharged state S_* is a certified internal edge $S \rightarrow_{\text{red}} S_*$ whose contract proves that the retained obstruction $I(S)$ transports to, implies, or realizes the retained obstruction $I(S_*)$. The contract also records compatibility of the scale, local window, representative, quotient data, and compactness topology. The target S_* must already be **closed** or **imported**, and the reduction relation must be acyclic, for example by decreasing a declared routing rank.

Reduction edges encode a common PDE move: a residual case is discharged by recognizing it as a configuration already excluded earlier in the proof. This internal proof edge is distinct from an imported analytic theorem. Its validity depends on the transported obstruction and on the compatibility data recorded in the two states.

Definition 7.7 (Local proof graph). A local proof graph is a finite directed graph

$$G = (V, E_{\text{route}}, E_{\text{red}})$$

with a root state. Route edges in E_{route} are labeled by local analytic capsules and construct descendant states. Reduction edges in E_{red} are obstruction-reduction edges to previously discharged states. Frozen terminal nodes carry one of the statuses

closed, imported, active-frontier.

Terminal means terminal for the route-edge relation. A closed terminal may carry a reduction edge as a closure dependency while remaining terminal for route construction. The status **active-frontier** is a partial-ledger status. It records a reachable frontier state for which the current graph has not yet supplied a closure capsule, an imported theorem, an obstruction-reduction edge, or a further stratification. Such a node is an unfinished ledger entry, not a closure used in the proof of the root claim. A completed proof graph has no reachable active-frontier nodes.

Remark 7.8 (Route is an edge type, not a closure status). The word *route* describes an edge action (E_{route}), not a closure verdict. Three distinct concepts must be kept separate:

1. **Construction-route edge.** A route edge in E_{route} creates or enters a new local capsule. It is not, by itself, a closure witness. A node that has only sent residuals to child states receives **active-frontier**, not **closed**.
2. **Obstruction-reduction edge.** A reduction edge in E_{red} closes the current state by proving that its retained obstruction transports to a previously discharged obstruction, with compatible coordinates and a rank or acyclicity witness. This IS a closure witness; the node receives **closed-by-reduction**.
3. **Router node / branch-exhaustion node.** A node whose job is to split cases into branches, where every outgoing branch terminates by **terminal-contradiction** or **obstruction-reduction**. Once all branches are closed the node receives **closed-by-branch-exhaustion**.

The complete set of allowed closure statuses in a certificate is:

{ **closed-by-direct-estimate**, **closed-by-import**, **closed-by-terminal-contradiction**, **closed-by-reduc**

The value **route** must not appear as a closure status in any completed proof object.

A node marked **closed** may be closed directly by a local terminal capsule or indirectly by a valid acyclic obstruction-reduction edge to a previously **closed** or **imported** node. In the latter case the reduction edge is part of the closure certificate and part of the dependency graph, but it is not a route edge that creates a new descendant state.

Definition 7.9 (Dependency cone). The dependency cone of an edge or node x is the full downstream subgraph of edges, nodes, and terminal claims whose certification uses x on at least one proof path.

A partially closed proof graph has a precise status. Its **active-frontier** nodes identify the remaining obligations for the chosen route-edge relation, and its dependency cones identify the part of the proof affected by any later invalidation.

8 Soundness, Partial Closure, and Fault Containment

Theorem 8.1 (Local proof graph soundness). *Let $G = (V, E_{\text{route}}, E_{\text{red}})$ be a finite rooted local proof graph. Assume that route construction is complete from the root, every route edge is justified by a valid local analytic capsule, every reduction edge is a valid obstruction-reduction edge, capsule interfaces compose along directed route paths, retained obstructions are preserved or transported to declared descendants, route construction terminates, every reachable frozen terminal is closed or imported, and the closure dependency relation, including E_{red} , is acyclic. Then no root counterexample branch exists.*

Proof. Suppose that a root branch exists. Route completeness supplies an outgoing success or failure route edge at each nonterminal state reached by the branch. Validity of the corresponding capsule produces the next branch with the declared output contract and transported obstruction. Termination gives a finite route path to a terminal node. The terminal is closed or imported, and the interface-composition hypothesis ensures that its input contract has been generated along the

path. If the terminal is imported, the imported theorem contradicts the transported obstruction under the recorded input contract. If it is directly closed, the local terminal capsule gives the contradiction. If it is closed by reduction, the valid reduction edge transports the terminal obstruction to a previously closed or imported obstruction. Acyclicity permits induction over the closure dependency relation, so the reduced target already contradicts its retained obstruction. The transported implication then contradicts the obstruction carried by the original branch. Hence the assumed root branch cannot exist. $\square$

Theorem 8.2 (Partial-closure ledger). *Let $G = (V, E_{\text{route}}, E_{\text{red}})$ be route-complete from the root, terminating, and labeled by valid composable capsules and valid obstruction-reduction edges that preserve or transport retained obstructions. Suppose the closure dependency relation, including E_{red} , is acyclic. Then every root branch reaches either a closed or imported terminal contradiction, possibly through reduction to a previously discharged obstruction, or a terminal marked **active-frontier** in the partial ledger. The realized **active-frontier** nodes are exactly the frozen frontier obligations for closing the root branch by this proof graph.*

Proof. The same path-realization argument used in [Theorem 8.1](#) sends every root branch to a terminal node. Closed and imported terminals contradict the transported obstruction, directly or through a valid acyclic reduction to a previously discharged obstruction. **active-frontier** nodes record unresolved frontier obligations and carry no contradiction. Conversely, a realized **active-frontier** node is reached by a route path from the root by definition, so it is a remaining obligation for this graph. $\square$

Proposition 8.3 (Local invalidation principle). *Assume certification status is propagated through dependency lists. If a capsule is invalidated, then only nodes, paths, and terminal claims in its dependency cone lose certification for that reason. Any branch with a closed or imported route path, including a reduction-closed path, avoiding the invalidated capsule remains closed by the unchanged subgraph.*

Proof. The dependency cone is defined as the downstream subgraph whose certification uses the invalidated capsule. Certification propagation can remove status only from claims whose dependency list contains that capsule. A route path avoiding the capsule has the same valid edges, the same transported obstruction, and the same terminal closure as before. It therefore remains certified unless another dependency is invalidated. $\square$

[Theorem 8.3](#) is the repair theorem. An invalid estimate has bounded scope: it invalidates a bounded region of the proof graph. Repair then consists of replacing the capsule, strengthening its input contract, adding a failure route, adding a valid reduction edge to a previously discharged obstruction, or returning the affected branch to the active frontier. If development stops before a replacement route is supplied, the partial ledger records the affected branch as an open obligation.

9 Linear PDE Stress Test: Local Stokes

9.1 Role in the paper

The Stokes section is a linear stress test for the proposed proof object and an explicit run of the construction algorithm in [Algorithm 1](#). The full analytic development is rendered in [Section C](#); the main text records how the local proof graph is produced.

The test begins from a standard failure mode in generated Stokes arguments. Curl removes pressure, vorticity solves a heat equation, and the generated proof infers local smoothness for the velocity. The inference loses the harmonic-gradient kernel. The construction algorithm repairs the statement by closing the vorticity-controlled component, promoting the remaining harmonic-gradient component to a named state, and choosing the next normalization from the structure of that residual. The graph in [Figure 1](#) is produced by this iteration.

Let

$$Q_R(z_0) = B_R(x_0) \times (t_0 - R^2, t_0)$$

and consider the local incompressible Stokes system

$$\partial_t u - \Delta u + \nabla p = f, \quad \operatorname{div} u = 0 \tag{1}$$

in distributions on $Q_R(z_0)$, with $u \in L^2_{\text{loc}}$ and smooth force f . The pressure is understood modulo functions of time. For a ball B , the local harmonic kernel $\mathcal{H}(B)$ consists of the $L^2(B)$ divergence-free and curl-free vector fields. On a ball these are harmonic-gradient velocity modes. The quotient representative is

$$u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u.$$

Theorem 9.1 (Stokes stress-test summary). *Let (u, p) solve (1) distributionally on Q_R , with $u \in L^2_{\text{loc}}$ and $f \in C^\infty(Q_R)$, and let*

$$Q_r(z_*) \subseteq B_\rho \times I_\rho \subseteq Q_R.$$

The local Stokes analysis in [Section C](#) gives the following three conclusions.

1. *Raw local regularity is false without a harmonic-kernel condition: time-dependent harmonic-gradient modes can be absorbed into the pressure and have zero vorticity.*
2. *After local harmonic-kernel normalization on $B_\rho \times I_\rho$, the representative $u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u$ is smooth on $Q_r(z_*)$, and the normalized pressure gradient $\nabla p^\perp$ is smooth there.*
3. *The same route certifies a chosen raw representative only when the removed harmonic-kernel component is zero or smooth on the smaller cylinder.*

Consequently a retained normalized singular obstruction cannot survive the local Stokes closure.

9.2 Running the construction algorithm

[Table 4](#) records the Stokes instance of [Algorithm 1](#). Each row is a construction step: a local theorem closes the component covered by its hypotheses, and the unclosed component determines the next state.

Table 4: Stokes instance of the local-closure algorithm.

Algorithmic operation	Stokes instance	State or edge produced
Retained obstruction	Start from a local Stokes branch on $Q_R(z_0)$ and a point z_* . For the normalized theorem the obstruction is $\mathfrak{s}^\perp(z_*)$: the representative $(I - \mathbf{P}_{\mathcal{H}(B_\rho)})u$ is singular in every smaller cylinder.	Root state with a transported normalized obstruction. For a raw representative, the corresponding obstruction is retained together with the harmonic-kernel compatibility condition.
Coordinate package	Choose $Q_r(z_*) \in B_\rho \times I_\rho \in Q_R$, the pressure representative modulo functions of time, the quotient by $\mathcal{H}(B_\rho)$, and the topology supplied by $u \in L^2_{\text{loc}}$ and $\text{curl } u \in H^{-1}_{\text{loc}}$.	Admissible local state $\Gamma(S)$ for vorticity heat smoothing, local pressure decomposition, and quotient div-curl recovery.
Local theorem matching	Apply curl to obtain $(\partial_t - \Delta) \text{curl } u = \text{curl } f$. Match this with local heat smoothing. Match $\Delta p = \text{div } f$ with the localized Calderon–Zygmund pressure row.	Imported local heat and pressure capsules with declared cylinder inputs and smaller-cylinder outputs.
Controlled closure	Heat smoothing closes $\text{curl } u$ on smaller cylinders. The local Calderon–Zygmund estimate closes the non-harmonic pressure component p_{CZ} . Quotient div-curl recovery closes the velocity component modulo $\mathcal{H}(B_\rho)$.	Closed non-harmonic local contribution and quotient representative $u^\perp$ as the active velocity variable.
Residual promotion	The curl-free and divergence-free velocity component $\mathbf{P}_{\mathcal{H}(B_\rho)}u$ remains. The pressure row also retains the harmonic pressure component $p_{\text{har}} + a(t)$.	Harmonic-kernel state and harmonic-pressure state, rather than a raw velocity-regularity conclusion.
Residual signature	The residual has local representative freedom in $\mathcal{H}(B_\rho)$, zero curl, zero divergence, and possible nonsmooth time dependence absorbed by the pressure representative.	The residual signature selects harmonic-kernel normalization and the corresponding pressure-representative transport.
Next normalization	Choose a harmonic potential Φ with $\nabla \Phi = \mathbf{P}_{\mathcal{H}(B_\rho)}u$, set $u^\perp = u - \nabla \Phi, \quad p^\perp = p + \partial_t \Phi,$ and keep the same local Stokes equation.	Normalized descendant state with $\mathbf{P}_{\mathcal{H}(B_\rho)}u^\perp = 0$.
Terminal closure	Apply quotient div-curl recovery and the pressure-normalization compatibility row on $Q_r(z_*)$.	Smoothness of $u^\perp$ and $\nabla p^\perp$, contradicting $\mathfrak{s}^\perp(z_*)$. A raw representative closes when the removed kernel is zero or smooth on the smaller cylinder.

Table 4: Stokes instance of the local-closure algorithm (continued).

Algorithmic operation	Stokes instance	State or edge produced
<i>Note.</i> The table constructs the graph from the analytic remainder left by each local closure. The harmonic-gradient component appears because the vorticity route has closed exactly the curl-controlled part and leaves the kernel outside its hypotheses. Its residual signature selects the quotient and pressure-normalization step.		

9.3 Capsule-level instantiation

Table 5 writes the same Stokes proof in capsule-interface form. Each row instantiates

$$\mathfrak{C} = (W, \Sigma, H, O, I, F, A),$$

where W is the local window, Σ records representative, quotient, scale, and topology data, H is the input contract, O is the output contract, I is the retained obstruction, F is the residual or failure route, and A is the audit condition. The analytic validation of these contracts is the standard PDE proof in Section C. The route and retained-obstruction columns give the row-level data for the graph edges displayed in Figure 1.

Table 5: Concrete Stokes capsules.

Capsule	Window W	State data Σ	Input contract H	Output O	Retained obstruction I	Route F	Audit A
Activity gate	$Q_\rho \in Q_R$, containing z_* .	Normalized branch state; raw representatives are allowed only with kernel compatibility.	A local Stokes branch (u, p, f) on Q_R , with $u \in L^2_{\text{loc}}$, $f \in C^\infty$, and pressure modulo functions of time.	Either a regular-window closure near z_* , or the vorticity-active state.	The normalized obstruction $\mathfrak{s}^\perp(z_*)$, or the raw obstruction together with kernel compatibility.	The two alternatives are exhaustive.	The obstruction is carried into the active state.
Vorticity heat capsule	$Q_r \in Q_\rho$.	$\text{curl } u \in H^{-1}_{\text{loc}}$, with smooth forcing topology on smaller cylinders.	The local Stokes equation, $u \in L^2_{\text{loc}}$, $f \in C^\infty$, and the constant-coefficient curl operation on Q_ρ .	$(\partial_t - \Delta) \text{curl } u = \text{curl } f$, and $\text{curl } u \in C^\infty(Q_r)$.	The obstruction is not discharged; only the curl-controlled component has been regularized.	Missing force regularity or missing weak curl class.	Heat smoothing consumes only data on Q_ρ .
Pressure atlas capsule	$B_r \in B_\sigma \in B_\rho$ and time interval I_ρ .	Pressure representative modulo functions of time; cutoff $\chi \in C_c^\infty(B_\rho)$.	The pressure equation $\Delta p = \text{div } f$, with $f \in C^\infty$.	Local splitting $p = p_{\text{CZ}} + p_{\text{har}} + a(t)$, where p_{CZ} is the smooth localized Newtonian potential of $\text{div}(\chi f)$.	The singular obstruction remains active; the harmonic pressure term is recorded as pressure-representative data.	Harmonic pressure state.	No whole-space pressure representative is consumed.
Quotient div-curl capsule	$B_r \times I_r \in B_\rho \times I_\rho$.	Quotient by $\mathcal{H}(B_\rho)$, with local projection $\mathbf{P}_{\mathcal{H}(B_\rho)}$.	$\text{div } u = 0$ and smooth $\text{curl } u$ on the local window.	Smooth quotient representative $u^\perp$ on the smaller cylinder.	The obstruction is transferred to the quotient representative together with the harmonic-kernel residual.	Harmonic-kernel residual.	The output is quotient-level until the kernel row closes.

Table 5: Concrete Stokes capsules (continued).

Capsule	Window W	State data Σ	Input contract H	Output O	Retained obstruction I	Route F	Audit A
Residual-promotion capsule	Same local cylinder.	Quotient data fixed; residual signature lies in the local kernel $\mathcal{H}(B_\rho)$.	The unresolved component $h = \mathbf{P}_{\mathcal{H}(B_\rho)}u$ after the vorticity and quotient div-curl capsules have closed their covered terms.	Harmonic-kernel state with zero curl, zero divergence, local representative freedom, and possible time dependence absorbed by pressure.	The retained obstruction is the harmonic-gradient residual that prevents raw velocity closure.	Raw representative compatibility.	The residual is named as a residual state before any velocity-regularity conclusion.
Harmonic-kernel normalization capsule	$B_\rho \times I_\rho$.	Zero-mean harmonic potential representative Φ , with $\nabla\Phi = h$.	$h = \mathbf{P}_{\mathcal{H}(B_\rho)}u \in L^2(I_\rho; L^2(B_\rho))$ and the harmonic lifting on the ball.	$u^\perp = u - \nabla\Phi$, $p^\perp = p + \partial_t\Phi$, with $\mathbf{P}_{\mathcal{H}(B_\rho)}u^\perp = 0$ and the same local Stokes equation.	The obstruction is transported to the normalized representative $(u^\perp, p^\perp)$.	Raw variables require zero-or-smooth kernel compatibility.	The pressure representative is transported with the velocity normalization.
Pressure normalization closure capsule	$Q_r(z_*)$.	Normalized pressure representative $p^\perp$.	Smooth $u^\perp$, smooth f , and the normalized local Stokes equation on the selected cylinder.	$\nabla p^\perp$ is smooth, since $\nabla p^\perp = f - \partial_t u^\perp + \Delta u^\perp$.	The retained normalized obstruction is ready to be consumed by terminal regularity.	Raw ∇p closes when the removed kernel is zero or smooth.	Only the pressure gradient is asserted, consistently with the time-only representative freedom.
Terminal contradiction capsule	$Q_r(z_*)$.	Retained-obstruction state $\mathfrak{s}^\perp(z_*)$.	Smoothness of $u^\perp$ and $\nabla p^\perp$ on $Q_r(z_*)$, produced by the previous capsules.	Contradiction with the retained normalized singular obstruction.	The obstruction $\mathfrak{s}^\perp(z_*)$ is consumed.	None in the completed normalized graph.	The terminal theorem consumes the obstruction assigned at the root.
<i>Note.</i> This table is the capsule-interface version of the proof. The appendix proves the analytic contracts in standard PDE form; the main text records the local windows, representatives, quotient data, retained obstruction, residual routes, and audit checks that the architecture requires.							

9.4 The resulting local proof graph

Figure 1 displays the graph produced by the steps in Table 4 and instantiated by the capsules in Table 5. From top to bottom, the graph records the successive outputs of Algorithm 1. Curl produces the vorticity heat route. The pressure row is kept as a local atlas. Quotient div-curl recovery reconstructs velocity modulo $\mathcal{H}(B_\rho)$. The dashed edge is the raw unnormalized failure; the solid route closes after local normalization and a pressure-normalization update.

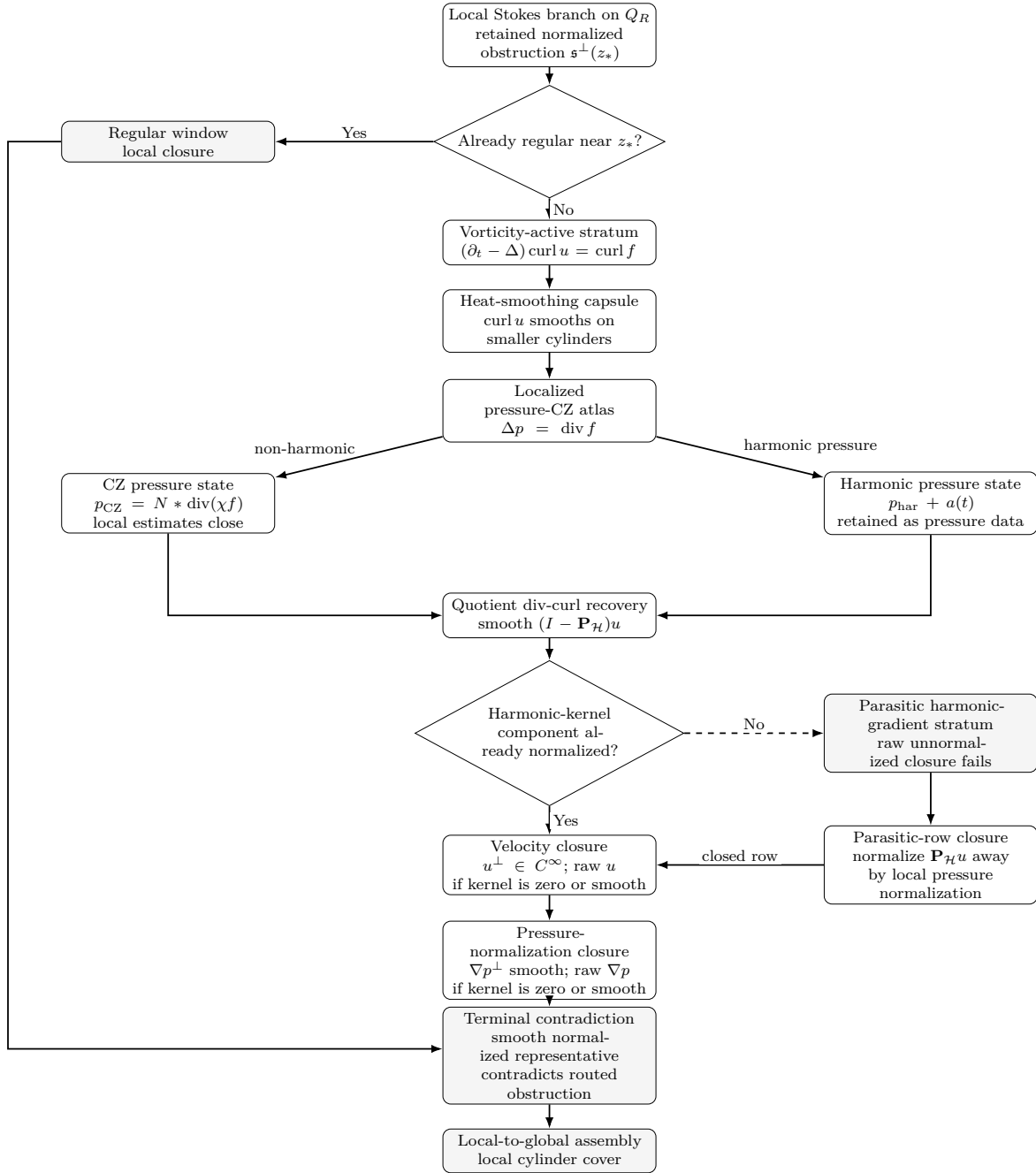


Figure 1: Stokes local proof graph produced by the construction algorithm. The analytic details are in [Section C](#). The dashed edge records the raw parasitic harmonic-gradient failure; the solid descendant route closes after local normalization.

9.5 Architectural consequences

The stress test isolates a concrete generated-PDE failure mode and shows how Algorithm 1 handles it. The vorticity route proves a quotient statement: it controls the curl-determined component and leaves a harmonic-gradient residual. The residual signature identifies the missing structure

as representative freedom in $\mathcal{H}(B_\rho)$, so the next local state is the harmonic-kernel normalization state. The invalid raw closure is thereby converted into a normalized closure theorem plus an explicit compatibility condition for raw representatives.

The example also shows the role of local PDE literature retrieval. Once the coordinate package is fixed, the required analytic inputs are standard local estimates: heat smoothing, localized Calderon–Zygmund pressure decomposition, quotient div-curl recovery, and harmonic-kernel normalization. The model-facing task is theorem matching inside that coordinate package: identify the correct local theorem, state its hypotheses, apply it to the covered component, and promote the residual term left outside its hypotheses.

The pressure and harmonic rows remain named states throughout the construction. They enter through explicit local contracts rather than through an undeclared whole-space projection, boundary condition, or global pressure convention.

10 Lean Certificate for the Finite Proof State

10.1 Certification boundary

The certified object is the construction ledger around the analytic estimates. An estimate enters the graph as a named capsule with a declared local window, input contract, output contract, pressure-representative or quotient status, topology, retained obstruction, and failure route. The certificate checks that the capsule is invoked only after its input contract has been produced, that its output is propagated with the declared topology and normalization, and that each terminal closure consumes the obstruction assigned to its branch. When a reduction edge is present, the same ledger records that its target is already closed or imported, its obstruction transport is declared, its analytic coordinates are compatible with the target state, and its dependency is acyclic.

Taxonomy completeness is also represented as a proof obligation. Each local state must declare its failure routes, and each terminal must declare its status. A regime discovered during audit is added as a new state, routed to an existing closure by a valid reduction edge, or kept on the active frontier until a closure, import, or reduction is supplied. In a partial snapshot, such a regime appears as an open obligation. When an imported estimate is corrected, strengthened, or rejected, branches outside its dependency cone keep their status.

10.2 Role of the backend

The Lean backend is a finite certificate for the Stokes proof state. The capsule interfaces in [Table 5](#) are the manuscript-level contracts; Lean represents their finite composition by a compressed state list, transition relation, local closure inputs, terminal capsules, and imported theorem boundaries. The analytic Stokes estimates remain named imported contracts. The certificate checks the finite structure around those imports; the PDE estimates themselves remain at the declared analytic boundary.

The checked Lean object mirrors the finite interface claims needed by the Stokes stress test. It contains fields for local-only assembly, finite state coverage, manuscript display-name alignment, transition soundness, terminal reachability, manuscript-label coverage, analytic-boundary coverage, pressure localization, harmonic pressure bookkeeping, quotient continuity, parasitic-row closure, and terminal capsule closure.

10.3 Checked finite route structure

The Lean certificate uses a compressed row list with eight states:

activity gate, vorticity heat, pressure-CZ normalization, quotient div-curl,
harmonic kernel, pressure normalization, empty residual state, terminal regularity.

The appendix and diagram display the finer analytic route states used for exposition; this compressed Lean list groups those states into certificate rows. The compression is the finite route abstraction checked by Lean; the analytic proof remains the Stokes argument in [Section C](#). Lean checks that the row list is complete, duplicate-free, has length eight, and carries the displayed manuscript names. It also checks the transition graph. The transitions are ranked:

$$\rho(\text{activity gate}) < \rho(\text{vorticity heat}) < \dots < \rho(\text{terminal regularity}),$$

and every declared transition strictly increases this rank. Thus the finite route graph is acyclic. Lean also proves that every listed state reaches terminal regularity in the reflexive-transitive closure of the declared transition relation. The current Stokes stress test has no nontrivial obstruction-reduction edges: its relevant branches close directly through local capsules or through imported analytic contracts. The general proof object nonetheless records reduction edges as separate closure dependencies.

10.4 Checked local assembly

The global Stokes conclusion in the backend is pointwise by construction. Its input is a local closure assignment for every domain point. Lean checks that these local assignments assemble pointwise into the conclusion

$$\forall z \in \Omega, \exists K \ni z \quad u \text{ is smooth on } K \quad \text{and} \quad \nabla p \text{ is smooth on } K.$$

This is the formal counterpart of the manuscript claim that global regularity is assembled from local closures only. The backend checks that the assembly stage uses no global heat kernel, global projection, or exterior pressure convention.

10.5 Checked pressure and harmonic rows

The pressure row is represented as a local atlas. Lean checks that the pressure-CZ source is localized before the CZ theorem is consumed, that the local pressure decomposition carries a harmonic pressure remainder, and that the harmonic remainder is retained as pressure-representative data. The harmonic velocity row is represented by a local projection state. Lean checks that the projected component is named, that the quotient carrier is available, and that the normalization step records the removed harmonic component.

The backend also packages the parasitic-row closure. The terminal capsule consumes a harmonic-kernel closure input: either the kernel is smooth or it has been normalized away. This matches the mathematical stress test above.

10.6 Analytic boundary

The Stokes backend treats the analytic estimates as named imported contracts. The currently declared boundary is listed in [Table 6](#).

Table 6: Imported analytic contracts in the Stokes finite certificate.

Imported contract	Role in the finite proof state
Harmonic projection	Produces the local quotient and harmonic-kernel state.
$L^2 \rightarrow H^{-1}$ curl bookkeeping	Places vorticity in the weak class needed for heat smoothing.
Smoothness of curled force	Supplies smooth forcing for the vorticity heat equation.
Local heat smoothing	Closes the vorticity regularity capsule.
Local Calderon–Zygmund pressure	Produces the localized pressure atlas and estimates for pcz .
Pressure-representative gradient invariance	Records that pressure is compared only modulo the allowed normalization.
Local div-curl recovery	Recovers velocity from local curl and divergence data modulo the harmonic kernel.
Quotient div-curl recovery	Recovers the quotient representative $(I - \mathbf{P}_{\mathcal{H}})u$.
Time-parametrized quotient recovery	Supplies time-derivative estimates for the quotient representative.

Note. Each row is an analytic import consumed by the finite proof-state certificate. The Lean layer checks where the import enters, which local state consumes it, and which terminal claims depend on it.

The axiom audit records the dependency boundary: the final validation certificate depends on these named analytic imports, on the geometric compact cylinder basis for open domains, and on standard Lean foundations. The final validation boundary lists these imports explicitly instead of introducing a single final regularity axiom.

10.7 Checked content

Lean checks the finite repair structure: state coverage, absence of duplicate states, strict transition ranking, terminal reachability, pointwise local-to-global assembly, localization before pressure-CZ consumption, explicit harmonic velocity and pressure-representative states, quotient continuity as a named contract, harmonic pressure normalization as a named row, parasitic-row closure as a terminal input, and exposure of the exact analytic boundary. This layer guards against global-estimate collapse. A replacement of any imported analytic estimate requires rechecking the capsules and dependency cones that consume that estimate.

11 Conclusion

This paper proposes a construction method for PDE proofs and an AI-facing architecture built around it. The proof is built by the loop in [Algorithm 1](#): choose an admissible local coordinate package, match a theorem from the local PDE literature, close the component covered by that theorem, promote the surviving remainder to a named state, and choose the next normalization, quotient, topology, or decomposition from the retained structure. Standard PDE tools enter as local capsules rather than as undeclared final shortcuts. Reduction to a previously discharged

obstruction is represented by a separate acyclic closure dependency. The output is a finite, typed, auditable, and repairable analytic proof graph.

The Stokes stress test shows the construction in a closed linear setting. The local vorticity and pressure estimates close the non-harmonic contribution. The remaining harmonic-gradient mode is promoted to a state, normalized by quotient div-curl recovery and pressure compatibility, and closed as the correct normalized theorem. The raw representative is recovered only under the zero-or-smooth compatibility condition. The Lean backend checks the finite certificate around this stress test and exposes the imported analytic boundary. AI-assisted PDE proof development then proceeds through local analytic capsules whose hypotheses, normalizations, compactness modes, obstructions, reduction edges, and failure routes are recorded during construction, audit, and repair. Each generated step must close a declared local component, import a named theorem, reduce to a discharged obstruction, or expose the next residual state.

A Implementation Capsule Template

This appendix renders the implementation-phase capsule template as a human-readable view of the canonical YAML certificate produced by the construction step. This phase proposes one local capsule, its theorem match, its residual routes, and its outgoing obstruction transports. It does not perform the adversarial review. The source is `docs/source/strategy_papers/implementation_capsule_template.md`.

```
<!--
IMPLEMENTATION CAPSULE TEMPLATE v1.0

Phase:
  Implementation / generation.

Parser rule:
1. Find the first fenced block with language 'yaml' immediately under
  '## MACHINE_RECORD'.
2. Parse that YAML block as the canonical implementation capsule.
3. Treat every Markdown section after the YAML block as human-readable rendering
  only.

Boundary:
  This template proposes a local capsule and its routed descendants. It does not
  decide whether the capsule passes adversarial review. The separate
  adversarial_evaluation_template.md consumes this record and issues the review
  verdict.
-->

# Implementation Capsule: [S_n] - [short name]

## MACHINE_RECORD

The YAML block below is canonical for the implementation phase. Markdown after
the YAML block may render the same information but must not add proof-bearing
claims.

```yaml
```

```

schema:
 name: "routed-capsule-implementation"
 version: "1.0.0"
 phase: "implementation"
 canonical_record: true
 parser_contract: "This YAML block is the source of truth for the proposed capsule."

rendering_contract:
 markdown_is_canonical: false
 yaml_is_source_of_truth: true
 semantic_sync_required: true
 proof_bearing_markdown_claim_policy:
 rule: "Every proof-bearing claim in Markdown must be represented in YAML."
 required_anchor_types:
 - hypothesis_id
 - theorem_id
 - requirement_id
 - residual_id
 - child_id
 - edge_id
 - obstruction_transport_id
 - closure_status

routing_semantics:
 route_is_not_closure_status: true
 construction_route_meaning: |-
 A construction-route edge creates or enters another local capsule.
 It is not, by itself, a closure witness.
 reduction_edge_meaning: |-
 An obstruction-reduction edge closes the current state only when the
 current retained obstruction transports to a previously discharged
 obstruction with a rank or acyclicity witness.
 closure_status_enum:
 - closed-by-direct-estimate
 - closed-by-import
 - closed-by-terminal-contradiction
 - closed-by-reduction
 - closed-by-branch-exhaustion
 - active-frontier
 edge_type_enum:
 - construction-route
 - obstruction-reduction
 - terminal-contradiction
 - import-dependency
 node_role_enum:
 - local-capsule
 - router
 - terminal
 - reduction

architecture:
 mode: "backward-contradiction"
 failed_control_semantics: "failed-control-becomes-routed-obstruction"

```



```

theorem_failure_semantics: "failed-hypothesis-becomes-child-state"
import_failure_semantics: "failed-import-becomes-child-state"
no_forward_tracking_claimed: true
no_forbidden_global_import_claimed: true

node:
 id: "[S_n]"
 name: "[short name]"
 rank: "[rank]"
 role: "[local-capsule/router/terminal/reduction]"
 parent_ids: ["[S_i]"]
 incoming_edge_label: "[why this node exists]"

retained_obstruction:
 id: "[I_n]"
 statement: |-
 [Exact obstruction carried by this node.]
 nontriviality_or_incompatibility: |-
 [Scalar lower bound, nonzero local quantity, incompatible topology, or profile property.]
 current_realization: |-
 [Where the obstruction is visible in this node.]

coordinate_package:
 z0:
 value: "[center/profile/component/null]"
 source: "[upstream ID]"
 if_missing_route_to: "[child ID or null]"
 scale:
 value: "[scale/null]"
 source: "[upstream ID]"
 if_missing_route_to: "[child ID or null]"
 W_core:
 value: "[core cylinder/subdomain]"
 source: "[upstream ID]"
 if_missing_route_to: "[child ID or null]"
 W_work:
 value: "[working cylinder/subdomain]"
 source: "[upstream ID]"
 if_missing_route_to: "[child ID or null]"
 representative:
 value: "[representative/gauge/normalization]"
 source: "[upstream ID]"
 if_missing_route_to: "[child ID or null]"
 quotient:
 value: "[quotient/kernel/modulo data]"
 source: "[upstream ID]"
 if_missing_route_to: "[child ID or null]"
 topology:
 available: "[available topology]"
 required_for_theorem: "[required topology]"
 verdict: "[sufficient/insufficient/not_applicable]"
 if_insufficient_route_to: "[child ID or null]"
 input_contract_ids:

```

```

- "[H1]"
- "[H2]"

input_contract:
 facts:
 - id: "H1"
 source: "[upstream node/import/convention]"
 statement: |-
 [Fact available before this node.]
 - id: "H2"
 source: "[upstream node/import/convention]"
 statement: |-
 [Fact available before this node.]

capsule:
 W: |-
 [Local cylinder or subdomain consumed.]
 Sigma:
 scale: "[scale]"
 representative: "[representative]"
 quotient: "[quotient]"
 topology: "[topology]"
 profile_variables: "[profile variables or null]"
 H:
 consumed_fact_ids: ["H1"]
 O_expected: |-
 [Exact covered output expected.]
 I_role: "[consumed/transported/retained/contradicted]"
 F_declared_routes:
 - route_id: "[route ID]"
 trigger: "[residual/failure condition]"
 target: "[child/reduction/terminal]"

theorem_match:
 exactly_one_theorem_or_identity_used: true
 theorem:
 id: "[T_local]"
 name: "[local theorem / identity / compactness tool]"
 source: "[citation or internal local identity]"
 allowed_import: true
 forbidden_global_import: false
 statement: |-
 [Precise local theorem used.]
 required_hypotheses:
 - id: "Req1"
 statement: |-
 [Required hypothesis.]
 matched_by: ["H1"]
 verdict: "[matched/missing/partially_matched]"
 if_missing_residual: "[R_missing_Req1 or null]"
 match_verdict: "[matched/partially_matched/not_matched]"
 scope_of_theorem: |-
 [Exactly what the theorem gives.]

```

```

theorem_does_not_give: |-
 [What must not be inferred.]

covered_contribution:
 id: "[COV1]"
 description: |-
 [Exact term/profile/component/topological condition/balance-law contribution closed by theorem.]
 output: |-
 [Exact local output.]
 nonclaims: |-
 [Explicitly list what is not closed here.]

failure_to_route_conversion:
 failed_items:
 - id: "[F1 or null]"
 kind: "[missing_hypothesis/unmatched_import/localized_error/topology_defect/representative_defect]"
 statement: |-
 [The failed item.]
 retained_obstruction_carried: |-
 [How I_n survives into this failure.]
 route_kind: "[child-local-capsule/profile-extraction/ranked-reduction/terminal-T1/terminal-T2/terminal-T3]"
 target: "[child ID / reduction target / terminal ID]"
 route_formula_or_map: |-
 [Exact obstruction transport map, scale normalization, profile extraction map, or contradiction.]
 rank_or_acyclicity_witness: |-
 [Required for reductions or recursive profile routes.]

profile_extraction:
 required: "[true/false]"
 trigger_residuals: "[R1]"
 extraction_if_required:
 compactness_mechanism: "[local compactness / Banach-Alaoglu / concentration compactness / profile compactness]"
 selected_sequence: |-
 [Sequence of centers, scales, local components, or defect points.]
 normalized_obstruction: |-
 [Fixed lower bound or nontrivial local quantity after rescaling.]
 topology: "[topology of convergence]"
 obstruction_transport_functional: |-
 [Continuous functional / balance identity / norm equivalence.]
 extracted_profile: "[Phi_* / u_* / component_* / limiting defect_*]"
 reduced_equation_or_reduced_object: |-
 [Equation/object satisfied by extracted profile.]
 next_route: "[terminal/reduction/child]"

obstruction_transport:
 id: "[OT1]"
 input_obstruction: " I_n "
 mechanism: "[continuous_functional/norm_equivalence/scale_normalization/profile_extraction/balance_law]"
 formula: |-
 [Exact formula proving the obstruction survives or is contradicted.]
 output_obstruction: "[I_{child} / terminal contradiction]"
 claimed_verdict: "[valid/invalid_creates_obstruction_loss_child]"

```

```

residuals:
 items:
 - id: "R1"
 description: |-
 [Residual after applying theorem to covered contribution.]
 why_not_covered: |-
 [Failed hypothesis / missing topology / unclassified profile / etc.]
 signature:
 scale: "[fixed/unstable/collapsing/multiscale]"
 locality: "[core_cylinder/working_cylinder/exterior_region/profile/topology]"
 representative_freedom: "[none/gauge/kernel/quotient]"
 topology: "[available vs needed]"
 defect_type: "[PDE defect type]"
 retained_obstruction: |-
 [Obstruction carried by this residual.]
 route_kind: "[child-local-capsule/profile-extraction/ranked-reduction/terminal-T1/terminal-T2/terminal-T3]"
 child_state: "[S_child or null]"
 blocker_language_used: false

child_states:
 items:
 - id: "[S_child]"
 name: "[short name]"
 produced_by_residual: "R1"
 rank: "[rank]"
 retained_obstruction: |-
 [Exact child obstruction.]
 input_contract:
 - id: "[child_H1]"
 statement: |-
 [Fact inherited by child.]
 coordinate_data_already_fixed:
 z0: "[value/null]"
 scale: "[value/null]"
 W_core: "[core cylinder or subdomain/null]"
 W_work: "[working cylinder or subdomain/null]"
 representative: "[value/null]"
 quotient: "[value/null]"
 topology: "[value/null]"
 required_next_move: "[local theorem/profile extraction/rank reduction/terminal contradiction]"
 do_not_solve_here: true

terminal_closure:
 is_terminal: "[true/false]"
 terminal_type: "[none/T1/T2/T3/named-obstruction]"
 terminal_profile_or_object: "[Phi_* / u_* / component_* / limiting defect_* / none]"
 contradiction_statement: |-
 [Exact contradiction, if terminal.]

closure_claim:
 node_role: "[local-capsule/router/terminal/reduction]"
 status: "[closed-by-direct-estimate/closed-by-import/closed-by-terminal-contradiction/closed-by-reduction]"
 reason: |-

```

```

 [Why this status is claimed.]
 if_status_closed_by_reduction:
 target_closed_node: "[S_closed]"
 target_closed_profile_or_obstruction: "[profile/obstruction already discharged]"
 obstruction_transport_map: |-
 [Formula showing I(current) realizes/implies/transport to I(target).]
 coordinate_compatibility: |-
 [Scale/domain/topology/representative compatibility.]
 rank_or_acyclicity_witness: |-
 [Strict rank decrease or acyclicity certificate.]
 if_status_closed_by_branch_exhaustion:
 branches:
 - branch_id: "[B1]"
 branch_output: "[profile/residual/defect produced]"
 edge_type: "[terminal-contradiction/obstruction-reduction/construction-route]"
 closure_target: "[terminal or already closed node]"
 obstruction_transport: |-
 [Map/formula.]
 branch_closure_status: "[closed/active-frontier]"

edges_out:
 items:
 - id: "[E1]"
 from: "[S_n]"
 to: "[S_child_or_terminal_or_reduction_target]"
 edge_type: "[construction-route/obstruction-reduction/terminal-contradiction/import-dependency]"
 produced_by: "[R1 / theorem output / terminal contradiction]"
 reason: |-
 [Why this edge exists.]
 obstruction_transport: |-
 [How the retained obstruction moves across this edge.]
 rank_rule: |-
 [Rank increases for child construction-route, or strict decrease/acyclic witness for obstruction]

implementation_self_report:
 solved_descendants: false
 claimed_global_target: false
 drew_full_DAG: false
 notes_for_reviewer: |-
 [Known delicate points, theorem-match uncertainties, or representative/topology issues to attack.]
'''

Human Rendering

After the YAML block, render only the same information in readable form. Use
this order:

1. node header;
2. retained obstruction;
3. coordinate package;
4. theorem match;
5. covered contribution;
6. obstruction transport;

```

```
7. residual routes;
8. child seeds;
9. terminal or closure claim;
10. edge output;
11. notes for reviewer.
```

Stop after this node. Do not perform the adversarial evaluation in this file.

## B Adversarial Evaluation Template

This appendix renders the review-phase template. The evaluator consumes an implementation capsule and produces an independent adversarial report: schema checks, theorem-match attacks, locality and representative attacks, residual routing checks, obstruction-transport checks, and a final accept, revise, or reject verdict. The source is `docs/source/strategy_papers/adversarial_evaluation_template.md`.

```
<!--
ADVERSARIAL EVALUATION TEMPLATE v1.0

Phase:
 Review / validation.

Parser rule:
1. Find the first fenced block with language 'yaml' immediately under
 '## MACHINE_RECORD'.
2. Parse that YAML block as the canonical adversarial review.
3. Treat every Markdown section after the YAML block as human-readable rendering
 only.

Boundary:
 This template consumes an implementation capsule. It does not generate a
 replacement proof node unless the verdict asks for a revision. Its job is to
 attack the implementation record and return precise defects, required
 reroutes, or acceptance.
-->

Adversarial Evaluation: [S_n] - [short name]

MACHINE_RECORD

The YAML block below is canonical for the review phase. It evaluates an
implementation capsule produced from 'implementation_capsule_template.md'.

''yaml
schema:
 name: "routed-capsule-adversarial-evaluation"
 version: "1.0.0"
 phase: "adversarial-evaluation"
 canonical_record: true
 parser_contract: "This YAML block is the source of truth for the review verdict."
```

```

review_target:
 implementation_schema_name: "routed-capsule-implementation"
 implementation_schema_version: "[version]"
 node_id: "[S_n]"
 implementation_record_hash_or_id: "[hash/path/id]"
 reviewer_id: "[review pass ID]"

verdict:
 status: "[accept/revise/reject]"
 summary: |-
 [One-paragraph review result.]
 accepted_closure_status: "[closure status or null]"
 must_return_to_implementation_phase: "[true/false]"

hard_reject_rules:
 items:
 - id: "ADV1"
 condition: "Uses Gronwall, bootstrap, continuity method, contraction mapping, or global-in-time"
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV2"
 condition: "Uses an operator-norm cascade or time-dependent radius to propagate a regime."
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV3"
 condition: "Treats a residual, failed theorem hypothesis, or failed import as a blocker instead"
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV4"
 condition: "Imports a forbidden global theorem, target theorem, black-box classification, or sh"
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV5"
 condition: "Claims closure status 'route' or otherwise treats route creation as closure."
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV6"
 condition: "Terminal closure is not T1, T2, T3, or named-obstruction."
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV7"
 condition: "Reduction edge lacks obstruction transport or rank/acyclicity witness."
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"
 - id: "ADV8"
 condition: "Markdown contains proof-bearing claims stronger than the YAML implementation record."
 triggered: false
 evidence: "[YAML path / Markdown anchor / none]"

schema_checks:
 yaml_present_and_parseable:
 pass: "[true/false]"
 evidence: "[path or parse error]"

```

```

required_fields_present:
 pass: "[true/false]"
 missing_fields:
 - "[field path]"
enum_values_valid:
 pass: "[true/false]"
 invalid_values:
 - path: "[field path]"
 value: "[invalid value]"
markdown_synced_to_yaml:
 pass: "[true/false]"
 stronger_markdown_claims:
 - "[claim and anchor]"

architecture_checks:
 backward_contradiction_mode:
 pass: "[true/false]"
 evidence: "[field path / reason]"
 no_forward_tracking:
 pass: "[true/false]"
 violations:
 - phrase_or_claim: "[claim]"
 location: "[field path / Markdown anchor]"
 required_conversion: "[child obstruction / profile extraction / reduction / terminal]"
 route_is_not_closure_status:
 pass: "[true/false]"
 evidence: "[closure field / edge field]"

theorem_match_attacks:
 exactly_one_local_theorem_or_identity:
 pass: "[true/false]"
 evidence: "[field path / reason]"
 hypotheses_match_input_contract:
 pass: "[true/false]"
 unmatched_requirements:
 - requirement_id: "[Req]"
 missing_or_partial_reason: "[reason]"
 required_reroute: "[residual ID / child state]"
 theorem_scope_not_overclaimed:
 pass: "[true/false]"
 overclaims:
 - claim: "[claim]"
 theorem_only_gives: "[scope]"
 required_revision: "[nonclaim / residual / child]"
 forbidden_import_attack:
 pass: "[true/false]"
 forbidden_imports:
 - theorem_id: "[T]"
 reason: "[why forbidden]"
 required_reroute: "[failed-import-obstruction]"

locality_scale_representative_attacks:
 locality_valid:

```



```

 pass: "[true/false]"
 defects:
 - "[window mismatch / unsupported enlargement / boundary term]"
scale_valid:
 pass: "[true/false]"
 defects:
 - "[scale mismatch / collapsing scale / missing normalization]"
representative_or_quotient_valid:
 pass: "[true/false]"
 defects:
 - "[gauge / pressure representative / kernel / quotient issue]"
topology_sufficient:
 pass: "[true/false]"
 defects:
 - "[available topology does not support theorem or obstruction transport]"

residual_routing_attacks:
 every_failed_hypothesis_routed:
 pass: "[true/false]"
 missing_routes:
 - "[failed item ID]"
 every_residual_has_route_kind:
 pass: "[true/false]"
 missing_routes:
 - "[residual ID]"
 no_blocker_language:
 pass: "[true/false]"
 blocker_claims:
 - "[claim]"
 child_states_seeded_but_not_solved:
 pass: "[true/false]"
 violations:
 - "[child state / descendant solved here]"

obstruction_transport_attacks:
 obstruction_carried_through_node:
 pass: "[true/false]"
 defects:
 - "[loss of lower bound / discontinuous functional / incompatible quotient]"
 edge_transport_explicit:
 pass: "[true/false]"
 missing_edges:
 - "[edge ID]"
 reduction_rank_or_acyclicity_valid:
 pass: "[true/false]"
 defects:
 - "[edge ID / rank issue]"
 terminal_contradiction_valid:
 pass: "[true/false]"
 defects:
 - "[terminal type / contradiction gap]"

required_revisions:

```

```

items:
 - id: "[REV1 or null]"
 severity: "[blocking/nonblocking]"
 target_yaml_path: "[implementation YAML path]"
 problem: |-
 [Precise defect.]
 required_change: |-
 [Exact implementation change needed.]
 acceptable_repair_kinds:
 - "[child-local-capsule/profile-extraction/ranked-reduction/terminal-T1/terminal-T2/terminal-T3]"

acceptance_certificate:
 all_hard_rejects_false: "[true/false]"
 all_required_schema_checks_pass: "[true/false]"
 all_proof_bearing_claims_anchored: "[true/false]"
 every_residual_routed_or_terminal_or_reduced: "[true/false]"
 obstruction_transport_valid: "[true/false]"
 final_review_verdict: "[accept/revise/reject]"
'''

Human Rendering

After the YAML block, render only the same review information in readable form.
Use this order:

1. review target;
2. verdict;
3. hard rejects;
4. schema checks;
5. theorem-match attacks;
6. locality, scale, representative, and topology attacks;
7. residual routing attacks;
8. obstruction transport attacks;
9. required revisions;
10. acceptance certificate.

Do not create a new implementation capsule in this file. If revision is needed,
return the implementation record to the implementation phase with the listed
required changes.

```

## C Local Stokes Stress Test: Analytic Details

### C.1 Statement of the Result

For  $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$  and  $R > 0$ , set

$$Q_R(z_0) = B_R(x_0) \times (t_0 - R^2, t_0).$$

When the center is clear we write  $Q_R$ . We consider the local incompressible Stokes system

$$\partial_t u - \Delta u + \nabla p = f, \quad \operatorname{div} u = 0 \quad (2)$$

in distributions on  $Q_R(z_0)$ . The pressure is normalized only modulo addition of a distribution depending only on time.

**Definition C.1** (Local harmonic kernel). For a ball  $B \Subset \mathbb{R}^3$ , set

$$\mathcal{H}(B) = \{h \in L^2(B; \mathbb{R}^3) : \operatorname{div} h = 0, \operatorname{curl} h = 0 \text{ in } B\}.$$

Equivalently, since  $B$  is simply connected, elements of  $\mathcal{H}(B)$  are gradients of harmonic potentials, up to the usual distributional identification. Let  $\mathbf{P}_{\mathcal{H}(B)}$  be the  $L^2(B)$ -orthogonal projection onto this closed subspace. A local Stokes solution is harmonic-kernel normalized on  $B \times I$  if

$$\mathbf{P}_{\mathcal{H}(B)}u(t) = 0 \quad \text{for a.e. } t \in I.$$

The field  $\mathbf{P}_{\mathcal{H}(B)}u$  is the harmonic-kernel, or parasitic, component of  $u$  on  $B \times I$ . The harmonic-kernel component is smooth on  $B \times I$  if it belongs to  $C^\infty(I; C^\infty(B'))$  for every  $B' \Subset B$ .

**Lemma C.2** (Properties of the local harmonic projection). *For every ball  $B$ , the space  $\mathcal{H}(B)$  is a closed subspace of  $L^2(B; \mathbb{R}^3)$ . Hence  $\mathbf{P}_{\mathcal{H}(B)}$  is a bounded linear operator on  $L^2(B; \mathbb{R}^3)$ . If  $u \in L^2(B \times I)$ , then*

$$t \mapsto \mathbf{P}_{\mathcal{H}(B)}u(t)$$

*is strongly measurable and belongs to  $L^2(I; L^2(B))$ . Moreover*

$$\operatorname{curl}(I - \mathbf{P}_{\mathcal{H}(B)})u = \operatorname{curl} u, \quad \operatorname{div}(I - \mathbf{P}_{\mathcal{H}(B)})u = \operatorname{div} u$$

*in distributions on  $B \times I$ .*

*Proof.* Let  $h_n \in \mathcal{H}(B)$  and  $h_n \rightarrow h$  in  $L^2(B)$ . Since divergence and curl are continuous from  $L^2(B)$  to  $H^{-1}(B)$ , the limits  $\operatorname{div} h = 0$  and  $\operatorname{curl} h = 0$  hold distributionally. Thus  $\mathcal{H}(B)$  is closed, and the Hilbert projection theorem gives the bounded orthogonal projection. Strong measurability and the  $L^2$  bound follow from bounded linearity. Finally,  $\mathbf{P}_{\mathcal{H}(B)}u(t)$  has zero divergence and zero curl for a.e.  $t$ , so subtracting it does not change either local distributional quantity.  $\square$

**Lemma C.3** (Local harmonic-kernel normalization). *Let  $B_\rho \times I_\rho \Subset Q_R(z_0)$ , and let  $(u, p, f)$  solve (2) in distributions there, with  $u \in L^2(B_\rho \times I_\rho)$ . Set*

$$h(t) = \mathbf{P}_{\mathcal{H}(B_\rho)}u(t).$$

*Then there is a distribution  $\Phi \in L^2(I_\rho; H^1(B_\rho))$ , harmonic in space for a.e.  $t$  and unique after imposing zero spatial mean, such that*

$$\nabla_x \Phi = h.$$

*The locally normalized representative*

$$u^\perp = u - h, \quad p^\perp = p + \partial_t \Phi$$

*satisfies*

$$\partial_t u^\perp - \Delta u^\perp + \nabla p^\perp = f, \quad \operatorname{div} u^\perp = 0$$

*in distributions on  $B_\rho \times I_\rho$ , and*

$$\mathbf{P}_{\mathcal{H}(B_\rho)}u^\perp(t) = 0 \quad \text{for a.e. } t \in I_\rho.$$

*Moreover  $\operatorname{curl} u^\perp = \operatorname{curl} u$ , and the local pressure equation  $\Delta p^\perp = \operatorname{div} f$  is unchanged.*

*Proof.* For a.e.  $t$ ,  $h(t) \in \mathcal{H}(B_\rho)$ . Since  $B_\rho$  is a ball, every element of  $\mathcal{H}(B_\rho)$  is the gradient of an  $H^1$  harmonic potential, unique modulo constants. Choosing the representative with zero spatial mean defines a bounded linear lifting

$$\mathcal{H}(B_\rho) \ni h \longmapsto \Phi_h \in H^1(B_\rho)/\mathbb{R}$$

by Poincaré's inequality. Applying this lifting to  $h(t)$  gives  $\Phi \in L^2(I_\rho; H^1(B_\rho))$ , with  $\nabla \Phi = h$  and  $\Delta \Phi = 0$  in distributions.

The divergence and curl identities follow from [Theorem C.2](#). Since  $h = \nabla \Phi$  and  $\Delta \Phi = 0$ , distributional differentiation gives

$$\partial_t(u - h) - \Delta(u - h) + \nabla(p + \partial_t \Phi) = \partial_t u - \Delta u + \nabla p = f,$$

because  $\partial_t \nabla \Phi = \nabla \partial_t \Phi$  and  $\Delta \nabla \Phi = \nabla \Delta \Phi = 0$ . Orthogonality gives

$$\mathbf{P}_{\mathcal{H}(B_\rho)}(u - h) = \mathbf{P}_{\mathcal{H}(B_\rho)}u - h = 0$$

for a.e.  $t$ . Finally, taking divergence of the normalized Stokes equation and using  $\operatorname{div} u^\perp = 0$  gives  $\Delta p^\perp = \operatorname{div} f$ , the same local pressure equation.  $\square$

*Remark C.4* (Why normalization is necessary). The regularity statement for a raw representative without a normalization hypothesis is false. If  $a \in L^2_{\operatorname{loc}}(I)$  is not smooth and  $a'(t)$  denotes its distributional derivative, then

$$u(x, t) = a(t)e_1, \quad p(x, t) = -a'(t)x_1, \quad f = 0$$

solves  $\partial_t u - \Delta u + \nabla p = 0$ ,  $\operatorname{div} u = 0$ , in distributions on every cylinder. Here  $\operatorname{curl} u = 0$ , so all vorticity estimates hold, but  $u$  is not smooth in time. This is the harmonic-gradient, or parasitic, velocity mode: it is curl-free and divergence-free, and its time dependence is absorbed by the pressure term  $\nabla p = -a'(t)e_1$ . Local normalization removes this mode; for a fixed raw representative, smoothness is a separate compatibility question for its harmonic-kernel component.

**Theorem C.5** (Local Stokes interior regularity). *Let  $(u, p)$  solve (2) in distributions on  $Q_R(z_0)$ , with*

$$u \in L^2_{\operatorname{loc}}(Q_R), \quad p \in \mathcal{D}'(Q_R), \quad f \in C^\infty(Q_R).$$

*Let  $B_r(y) \times I_r \Subset B_\rho(y) \times I_\rho \Subset Q_R(z_0)$ , and write*

$$u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u$$

*on  $B_\rho \times I_\rho$ . Then  $u^\perp$  is smooth on  $B_r \times I_r$ . More precisely, for every  $j, k \geq 0$  there are an integer  $N_{j,k}$  and a constant  $C_{j,k,r,\rho,I_r,I_\rho}$ , depending only on the nested cylinders and derivative orders, such that*

$$\|\partial_t^j \nabla^k u^\perp\|_{L^2(B_r \times I_r)} \leq C_{j,k,r,\rho,I_r,I_\rho} \left( \|u\|_{L^2(B_\rho \times I_\rho)} + \sum_{\ell+|\alpha| \leq N_{j,k}} \|\partial_t^\ell \nabla^\alpha f\|_{L^2(B_\rho \times I_\rho)} \right). \quad (3)$$

*If, in addition, the harmonic-kernel component  $\mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on  $B_r \times I_r$ , then*

$$u \in C^\infty(B_r \times I_r), \quad \nabla p \in C^\infty(B_r \times I_r).$$

*In particular, full smoothness holds when the solution is harmonic-kernel normalized on  $B_\rho \times I_\rho$ .*

**Theorem C.6** (Global regularity assembled from local estimates). *Let  $\mathcal{U} \subset \mathbb{R}^3 \times \mathbb{R}$  be open, and let  $(u, p, f)$  solve (2) in distributions on  $\mathcal{U}$ , with*

$$u \in L_{\text{loc}}^2(\mathcal{U}), \quad p \in \mathcal{D}'(\mathcal{U}), \quad f \in C^\infty(\mathcal{U}).$$

*Assume that for every  $z \in \mathcal{U}$  there is a nested cylinder*

$$z \in B_r \times I_r \Subset B_\rho \times I_\rho \Subset \mathcal{U}$$

*on which the harmonic-kernel component  $\mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on  $B_r \times I_r$ , or the representative under consideration is already harmonic-kernel normalized there. Then*

$$u \in C^\infty(\mathcal{U}), \quad \nabla p \in C^\infty(\mathcal{U}).$$

*The proof uses only the local estimates of Theorem C.5 on the cylinders in this cover.*

*Proof.* For each  $z \in \mathcal{U}$ , apply Theorem C.5 on the cylinder supplied by the hypothesis. This gives a neighborhood of  $z$  on which  $u$  and  $\nabla p$  are smooth. Since smoothness is local, these neighborhoods cover  $\mathcal{U}$ , and the conclusion follows. If one wants a compact-set estimate, cover a compact  $K \Subset \mathcal{U}$  by finitely many such cylinders and take the maximum of the corresponding local constants. No estimate ever uses data outside its selected larger cylinder.  $\square$

**Theorem C.7** (Global regularity for the normalized representative). *Let  $(u, p, f)$  solve (2) in distributions on  $\mathcal{U}$ , with*

$$u \in L_{\text{loc}}^2(\mathcal{U}), \quad p \in \mathcal{D}'(\mathcal{U}), \quad f \in C^\infty(\mathcal{U}).$$

*For every  $z \in \mathcal{U}$ , choose nested cylinders*

$$z \in B_r \times I_r \Subset B_\rho \times I_\rho \Subset \mathcal{U}$$

*and form the normalized representative  $(u^\perp, p^\perp)$  on the larger cylinder by Theorem C.3. Then*

$$u^\perp \in C^\infty(B_r \times I_r), \quad \nabla p^\perp \in C^\infty(B_r \times I_r).$$

*If, in addition, the original harmonic-kernel component  $\mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on the smaller cylinders in such a cover, or the representative under consideration is already harmonic-kernel normalized there, then the original representative satisfies*

$$u \in C^\infty(\mathcal{U}), \quad \nabla p \in C^\infty(\mathcal{U}).$$

*On each compact  $K \Subset \mathcal{U}$ , all estimates are obtained from finitely many such local cylinders.*

*Proof.* The normalized representative satisfies the same local Stokes system with smooth force  $f$ , is divergence-free, and has zero harmonic-kernel component on  $B_\rho \times I_\rho$  by Theorem C.3. Applying Theorem C.5 on the nested cylinders gives smoothness of  $u^\perp$  and  $\nabla p^\perp$  on  $B_r \times I_r$ . If the original kernel component is smooth or zero on the same smaller cylinder, then

$$u = u^\perp + \mathbf{P}_{\mathcal{H}(B_\rho)}u$$

is smooth there. The pressure relation  $p^\perp = p + \partial_t \Phi$  gives

$$\nabla p = \nabla p^\perp - \partial_t \mathbf{P}_{\mathcal{H}(B_\rho)}u$$

on the smaller cylinder, hence  $\nabla p$  is smooth whenever the original kernel component is smooth or zero. A finite subcover of each compact  $K \Subset \mathcal{U}$  gives compact-set estimates and uses no data outside the selected larger cylinders.  $\square$

*Remark C.8.* Global Stokes regularity is assembled pointwise from local estimates: no global heat kernel, whole-space Leray projection, or boundary condition at infinity is used.

*Remark C.9* (Local Calderón–Zygmund pressure decomposition). Taking curl in the Stokes equation removes pressure from the vorticity argument, and velocity can be recovered from the local pair  $\operatorname{div} u = 0$ ,  $\operatorname{curl} u = \omega$  modulo the explicitly identified harmonic-kernel component. Pressure is handled separately: one chooses a local representative, splits it into a localized Calderón–Zygmund part and a harmonic pressure remainder, and shows the harmonic remainder is precisely the term carrying any parasitic time dependence. In the Stokes case, the source  $\Delta p = \operatorname{div} f$  is linear, so the Calderón–Zygmund part closes directly; the only pressure term requiring separate attention is the harmonic component, which is absorbed by the harmonic-kernel normalization. After the velocity is smoothed, the pressure gradient is recovered by the formula  $\nabla p = f - \partial_t u + \Delta u$ .

*Remark C.10.* The smoothness assumption on  $f$  is chosen to keep the statement clean. The same proof gives the standard Sobolev or parabolic Hölder interior estimates when  $f$  is placed in those local classes.

## C.2 Candidate Singularities and Proof Steps

In this subsection we record the proof steps underlying [Theorem C.5](#) in a form that makes explicit which local argument closes which term. This organization makes the harmonic-kernel compatibility condition visible at each step.

**Proposition C.11** (Ordered regularity alternatives). *Let  $(u, p, f)$  solve (2) on  $Q_R(z_0)$  with  $u \in L^2_{\operatorname{loc}}(Q_R)$  and  $f \in C^\infty(Q_R)$ . Let  $z_* \in Q_{R/4}(z_0)$ . The following cases are exhaustive and mutually exclusive, and each closes by a local argument:*

1. *The representative under consideration is already smooth near  $z_*$ .*
2. *The vorticity  $\omega = \operatorname{curl} u$  is the active variable; it satisfies the local heat equation  $(\partial_t - \Delta)\omega = \operatorname{curl} f$  and is smooth on smaller cylinders.*
3. *After vorticity smoothing, the remaining curl-free component of the velocity field lies in the local harmonic-kernel class; it is smooth in space by elliptic regularity.*
4. *The harmonic-kernel component is handled by local normalization; for a raw representative, the additional condition is that this component be zero or smooth on the smaller cylinder.*
5. *The Calderón–Zygmund pressure argument controls the non-harmonic part of the pressure locally; its harmonic part is a spatial-harmonic distribution in time, which is absorbed by the harmonic-kernel normalization.*
6. *Once the normalized representative is smooth, smoothness of the pressure gradient follows from the Stokes equation itself.*

*All steps use only data on nested cylinders, and the argument at each step uses no data outside the selected larger cylinder.*

*Proof.* If the representative is already smooth near  $z_*$ , the proof is complete. Otherwise,  $u \in L^2_{\operatorname{loc}}$  gives  $\omega = \operatorname{curl} u \in H^{-1}_{\operatorname{loc}}$ , so the curl equation is a legitimate local consequence of the Stokes

system. Vorticity smoothing controls the non-harmonic spatial part of  $u$ . If the curl contribution vanishes on a time slice, the identity

$$-\Delta u = \operatorname{curl} \operatorname{curl} u - \nabla \operatorname{div} u$$

places that component in the harmonic-kernel class, which is locally smooth by elliptic regularity in space. If the solution is not already harmonic-kernel normalized, [Theorem C.3](#) normalizes the local representative and produces a Stokes solution with zero harmonic-kernel component. The example in [Theorem C.4](#) shows that raw full regularity requires the removed kernel to be separately zero or smooth. After the velocity is recovered, the Stokes equation determines only  $\nabla p$ , so the final step is the pressure normalization  $p \sim p + a(t)$ . These cases exhaust all terms in (2): curl removes pressure from the vorticity argument, Calderón–Zygmund estimates control the localized non-harmonic pressure, div-curl recovers velocity, and the equation recovers the final pressure gradient.  $\square$

**Proposition C.12** (Reduction to the vorticity equation). *Let  $(u, p, f)$  solve (2) on  $Q_R(z_0)$ , with  $u \in L^2_{\operatorname{loc}}(Q_R)$  and  $f \in C^\infty(Q_R)$ , and let  $z_* \in Q_{R/4}(z_0)$ . Either  $u$  is smooth near  $z_*$ , or  $\omega = \operatorname{curl} u$  satisfies the local heat equation*

$$(\partial_t - \Delta)\omega = \operatorname{curl} f$$

on some cylinder  $Q_\rho$  with  $z_* \in Q_\rho \Subset Q_R$ .

*Proof.* If  $u$  is smooth near  $z_*$ , there is nothing to prove. Otherwise,  $u \in L^2_{\operatorname{loc}}(Q_R)$  gives  $\omega = \operatorname{curl} u \in H^{-1}_{\operatorname{loc}}(Q_R)$ . Taking curl in the Stokes equation is a local distributional operation and requires no data outside  $Q_R$ . This gives the stated local heat equation on any cylinder  $Q_\rho \Subset Q_R$  containing  $z_*$ .  $\square$

### C.3 Vorticity Smoothing

**Lemma C.13** (Vorticity equation). *Let  $(u, p, f)$  solve (2) in distributions on  $Q_R$ . Then  $\omega = \operatorname{curl} u$  satisfies*

$$(\partial_t - \Delta)\omega = \operatorname{curl} f \tag{4}$$

in distributions on  $Q_R$ .

*Proof.* Apply curl to (2). Constant-coefficient differentiation commutes with  $\partial_t$  and  $\Delta$ , and  $\operatorname{curl} \nabla p = 0$ . The result is (4).  $\square$

**Lemma C.14** (Interior heat smoothing). *Let  $w \in H^{-1}_{\operatorname{loc}}(Q_R)$  solve*

$$(\partial_t - \Delta)w = g$$

in distributions on  $Q_R$ , with  $g \in C^\infty(Q_R)$ . If  $0 < r < \rho < R$ , then  $w \in C^\infty(Q_r)$ . For every  $j, k \geq 0$ ,

$$\|\partial_t^j \nabla^k w\|_{L^2(Q_r)} \leq C_{j,k,r,\rho} \left( \|w\|_{H^{-1}(Q_\rho)} + \sum_{\ell+|\alpha| \leq M_{j,k}} \|\partial_t^\ell \nabla^\alpha g\|_{L^2(Q_\rho)} \right). \tag{5}$$

*Proof.* Choose nested cylinders

$$Q_r \Subset Q_{\sigma_0} \Subset Q_{\sigma_1} \Subset Q_\rho.$$

We use the standard local parametrix form of heat hypoellipticity. In anisotropic Sobolev notation, where one time derivative has parabolic order two, the estimate says the following. For cutoffs  $\chi_0, \chi_1 \in C_c^\infty(Q_{\sigma_1})$ , with  $\chi_0 \equiv 1$  on  $Q_{\sigma_0}$  and  $\chi_1 \equiv 1$  on  $\text{supp } \chi_0$ , and for every integer  $s \geq -1$ ,

$$\|\chi_0 w\|_{H_{\text{par}}^{s+2, s/2+1}(Q_{\sigma_1})} \leq C_{s, \chi_0, \chi_1} \left( \|\chi_1 (\partial_t - \Delta) w\|_{H_{\text{par}}^{s, s/2}(Q_{\sigma_1})} + \|\chi_1 w\|_{H^{-1}(Q_{\sigma_1})} \right). \quad (6)$$

This estimate is local: it is obtained by applying a compactly supported constant-coefficient parametrix for  $\partial_t - \Delta$  to  $\chi_0 w$ . The commutator  $[\partial_t - \Delta, \chi_0]w$  is supported in  $\{\chi_1 = 1\} \setminus Q_{\sigma_0}$ . At the first step it is measured in a negative Sobolev norm controlled by  $\|\chi_1 w\|_{H^{-1}}$ ; after each bootstrap step the same estimate is applied on a smaller intermediate cylinder, so the commutator only uses regularity already gained on the previous cylinder. No value of  $w$  outside  $Q_{\sigma_1}$  enters.

Apply (6) first with  $s = -1$  and  $(\partial_t - \Delta)w = g$ . Since  $g$  is smooth, this raises the Sobolev regularity of  $w$  on  $Q_{\sigma_0}$ . Repeating the argument on a finite chain of nested cylinders gives, for every parabolic order  $N$ ,

$$\|w\|_{H_{\text{par}}^{N, N/2}(Q_r)} \leq C_{N, r, \rho} \left( \|w\|_{H^{-1}(Q_\rho)} + \sum_{\ell+|\alpha| \leq M_N} \|\partial_t^\ell \nabla^\alpha g\|_{L^2(Q_\rho)} \right).$$

Choosing  $N$  large enough to dominate the parabolic order  $2j + k$  and then using local parabolic Sobolev embedding gives (5) and smoothness.  $\square$

**Corollary C.15** (Vorticity smoothing). *Under the hypotheses of Theorem C.5,  $\omega = \text{curl } u$  is smooth on every compactly contained subcylinder  $Q' \Subset Q_R$ . Its derivatives satisfy local estimates on nested cylinders  $Q' \Subset Q'' \Subset Q_R$  in terms of local norms of  $u$  and local derivatives of  $f$  on  $Q''$ .*

*Proof.* Since  $u \in L_{\text{loc}}^2$ ,  $\omega = \text{curl } u \in H_{\text{loc}}^{-1}$ . Apply Theorems C.13 and C.14 with  $g = \text{curl } f$ .  $\square$

## C.4 Local Pressure Normalization and Calderón–Zygmund Decomposition

**Lemma C.16** (Local pressure equation). *Let  $(u, p, f)$  solve (2) in distributions on  $Q_R$ . Then*

$$\Delta p = \text{div } f \quad (7)$$

*in distributions on  $Q_R$ .*

*Proof.* Apply divergence to the Stokes equation. Since  $\text{div } u = 0$ ,

$$\partial_t \text{div } u - \Delta \text{div } u + \Delta p = \text{div } f$$

reduces to (7).  $\square$

**Lemma C.17** (Localized Calderón–Zygmund pressure decomposition). *Let  $B_r \times I_r \Subset B_\sigma \times I_\sigma \Subset B_\rho \times I_\rho \Subset Q_R$ , and let  $(u, p, f)$  solve (2). Choose  $\chi \in C_c^\infty(B_\rho)$  with  $\chi \equiv 1$  on  $B_\sigma$ . Let  $N$  be the Newtonian potential with  $\Delta N = \delta_0$ . On  $B_\sigma \times I_\sigma$  there is a distributional decomposition, unique up to a function of time,*

$$p = p_{\text{CZ}} + p_{\text{har}} + a(t),$$

*where*

$$p_{\text{CZ}}(t) = N * \text{div}(\chi f(t))$$



and

$$\Delta_x p_{\text{har}} = 0 \quad \text{in } \mathcal{D}'(B_\sigma \times I_\sigma).$$

Equivalently,  $p_{\text{har}}$  is a distribution in time with values in spatial harmonic distributions on  $B_\sigma$ . The term  $a(t)$  denotes the remaining distribution that is constant in space. For every  $j, k \geq 0$ ,

$$\|\partial_t^j \nabla^{k+1} p_{\text{CZ}}\|_{L^2(B_r \times I_r)} \leq C_{j,k,r,\sigma,\rho,I_r,I_\rho} \sum_{\ell+|\alpha| \leq j+k} \|\partial_t^\ell \nabla^\alpha f\|_{L^2(B_\rho \times I_\rho)}. \quad (8)$$

Thus the non-harmonic pressure gradient is locally smooth and quantitatively controlled by the selected cylinder data only.

*Proof.* The identity (7) gives

$$\Delta(p - p_{\text{CZ}}) = \operatorname{div} f - \operatorname{div}(\chi f) = \operatorname{div}((1 - \chi)f).$$

The right side vanishes on  $B_\sigma \times I_\sigma$ , because  $\chi = 1$  there. Hence  $p_{\text{har}} = p - p_{\text{CZ}}$ , modulo a distribution depending only on time, is harmonic in  $x$  on  $B_\sigma \times I_\sigma$ . The time-only distribution is harmless for all later statements, which use  $\nabla p$ .

For the estimate, write

$$\nabla p_{\text{CZ}} = \nabla \nabla N * (\chi f),$$

componentwise, with the sign fixed by the convention  $\Delta N = \delta_0$ . The kernels  $\partial_i \partial_j N$  are Calderón–Zygmund kernels. The  $L^2$  Calderón–Zygmund bound applied to the compactly supported field  $\chi f$ , followed by the Leibniz rule for derivatives falling on  $\chi$  and  $f$ , gives (8). The cutoff is fixed inside  $B_\rho$ , so the estimate consumes no exterior pressure value, no boundary condition, and no whole-space pressure reconstruction.  $\square$

**Proposition C.18** (Pressure decomposition and harmonic remainder). *On every nested cylinder, the pressure gradient splits as*

$$\nabla p = \nabla p_{\text{CZ}} + \nabla p_{\text{har}}.$$

The Calderón–Zygmund part  $\nabla p_{\text{CZ}}$  is smooth and quantitatively controlled by [Theorem C.17](#). The harmonic pressure gradient  $\nabla p_{\text{har}}$  lies in the same local harmonic-gradient class as the parasitic velocity mode of [Theorem C.1](#): it can carry nonsmooth time dependence. Harmonic-kernel normalization absorbs this harmonic pressure term; for a raw representative, smoothness of the harmonic-kernel component is the extra condition needed to conclude smoothness of  $\nabla p$ .

*Proof.* The decomposition is [Theorem C.17](#). Since  $\nabla p_{\text{har}}$  is divergence-free and curl-free in the spatial variables, it lies in the same local harmonic-gradient class that appears in [Theorem C.1](#). The example in [Theorem C.4](#) shows that this term can carry nonsmooth time dependence. Hence it must be identified and handled through the harmonic-kernel normalization, not treated as automatically smooth.  $\square$

## C.5 Velocity Recovery

**Lemma C.19** (Local div-curl estimate). *Let  $B_r \Subset B_\rho \subset \mathbb{R}^3$ . Suppose  $v \in L^2(B_\rho)$ ,  $\operatorname{div} v = 0$  in  $B_\rho$ , and  $\operatorname{curl} v$  is smooth in  $B_\rho$ . Then  $v$  is smooth in  $B_r$ . For every  $m \geq 0$ ,*

$$\|\nabla^{m+1} v\|_{L^2(B_r)} \leq C_{m,r,\rho} \left( \|v\|_{L^2(B_\rho)} + \sum_{|\alpha| \leq m} \|\nabla^\alpha \operatorname{curl} v\|_{L^2(B_\rho)} \right). \quad (9)$$

*Proof.* The vector identity

$$-\Delta v = \operatorname{curl} \operatorname{curl} v - \nabla \operatorname{div} v$$

gives  $-\Delta v = \operatorname{curl} \operatorname{curl} v$  in  $B_\rho$ . Interior elliptic estimates for the vector Poisson equation on  $B_r \Subset B_\rho$  imply (9). Iteration and Sobolev embedding give smoothness. If  $\operatorname{curl} v = 0$ , this says simply that  $v$  is harmonic and therefore smooth; this is the local harmonic-kernel case.  $\square$

**Lemma C.20** (Local quotient div-curl isomorphism). *Let  $B_r \Subset B_\rho$  be concentric balls. There is a local linear recovery operator, defined on the range of curls of divergence-free  $L^2(B_\rho)$  fields,*

$$\mathcal{T}_{B_\rho} \omega = v,$$

*characterized by*

$$\operatorname{div} v = 0, \quad \operatorname{curl} v = \omega, \quad v \perp \mathcal{H}(B_\rho) \text{ in } L^2(B_\rho).$$

*If  $u \in L^2(B_\rho)$ ,  $\operatorname{div} u = 0$ , and  $\omega = \operatorname{curl} u$ , then*

$$\mathcal{T}_{B_\rho} \omega = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u.$$

*For smooth divergence-free curl data, the same operator is equivalently the orthogonal Hodge representative of the local div-curl system below. For every  $m \geq 0$ ,*

$$\|\nabla^{m+1} \mathcal{T}_{B_\rho} \omega\|_{L^2(B_r)} \leq C_{m,r,\rho} \left( \|\mathcal{T}_{B_\rho} \omega\|_{L^2(B_\rho)} + \sum_{|\alpha| \leq m} \|\nabla^\alpha \omega\|_{L^2(B_\rho)} \right). \quad (10)$$

*The constants depend only on the nested balls and derivative order.*

*Proof.* Existence for  $\omega = \operatorname{curl} u$  is obtained by setting

$$v = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u.$$

**Theorem C.2** gives  $\operatorname{div} v = 0$  and  $\operatorname{curl} v = \omega$ , and the definition of the projection gives  $v \perp \mathcal{H}(B_\rho)$ . Conversely, if  $v_1$  and  $v_2$  satisfy the three displayed conditions, then  $v_1 - v_2 \in \mathcal{H}(B_\rho)$  and  $v_1 - v_2 \perp \mathcal{H}(B_\rho)$ , hence  $v_1 = v_2$ . Thus the quotient representative is uniquely determined by the local curl data.

For smooth divergence-free  $\omega$ , the standard local Hodge construction on the ball gives the same orthogonal representative; equivalently, solve the div-curl system modulo  $\mathcal{H}(B_\rho)$ . Interior estimates are insensitive to the boundary realization of this representative. The estimate is **Theorem C.19** applied to the representative. The quotient condition is essential: without  $v \perp \mathcal{H}(B_\rho)$ , the curl data would not determine the velocity.  $\square$

**Lemma C.21** (Sobolev continuity of quotient recovery). *Let  $B_r \Subset B_\rho$  be concentric balls. For every  $m \geq 0$ , the quotient recovery map  $\mathcal{T}_{B_\rho}$  from **Theorem C.20** is continuous from compatible curl data in  $H^m(B_\rho)$  to  $H^{m+1}(B_r)$ . Through the smooth Hodge extension, the same continuity holds for smooth divergence-free curl data. More precisely, if  $\omega = \operatorname{curl} u$  for some divergence-free  $u \in L^2(B_\rho)$ , then*

$$\|\mathcal{T}_{B_\rho} \omega\|_{H^{m+1}(B_r)} \leq C_{m,r,\rho} \left( \|\mathcal{T}_{B_\rho} \omega\|_{L^2(B_\rho)} + \|\omega\|_{H^m(B_\rho)} \right).$$

*Consequently, for any distribution  $W$  in time with values in compatible curl data, distributional time differentiation commutes with  $\mathcal{T}_{B_\rho}$ :*

$$\partial_t \mathcal{T}_{B_\rho} W = \mathcal{T}_{B_\rho} (\partial_t W)$$

*as distributions on smaller cylinders whenever both sides are tested on  $B_r \Subset B_\rho$ .*

*Proof.* The displayed Sobolev estimate is the sum of Equation (10) for orders  $0, \dots, m$ , plus the  $L^2$  term. Thus  $\mathcal{T}_{B_\rho}$  is a fixed continuous linear map between the indicated local Sobolev spaces on its compatible-data domain, and the same interior estimate applies to the smooth Hodge representative of smooth divergence-free curl data. Continuity and time-independence imply commutation with distributional time derivatives: for every temporal test function  $\eta$  and spatial test functional  $\Phi$  supported in  $B_r$ ,

$$\langle \partial_t \mathcal{T}_{B_\rho} W, \eta \Phi \rangle = -\langle \mathcal{T}_{B_\rho} W, \eta' \Phi \rangle = -\langle W, \eta' (\mathcal{T}_{B_\rho})^* \Phi \rangle = \langle \mathcal{T}_{B_\rho} (\partial_t W), \eta \Phi \rangle.$$

This proves the distributional commutation statement.  $\square$

**Lemma C.22** (Time-parametrized quotient recovery). *Let  $B_r \times I_r \Subset B_\rho \times I_\rho$ . Suppose*

$$u \in L^2(B_\rho \times I_\rho), \quad \operatorname{div} u = 0, \quad \omega = \operatorname{curl} u \in C^\infty(B_\rho \times I_\rho).$$

*Then  $u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u$  is smooth on  $B_r \times I_r$ . For all  $j, k \geq 0$ ,*

$$\|\partial_t^j \nabla^k u^\perp\|_{L^2(B_r \times I_r)} \leq C_{j,k,r,\rho,I_r,I_\rho} \left( \|u^\perp\|_{L^2(B_\rho \times I_\rho)} + \sum_{\ell \leq j, |\alpha| \leq k+j} \|\partial_t^\ell \nabla^\alpha \omega\|_{L^2(B_\rho \times I_\rho)} \right). \quad (11)$$

*Proof.* For a.e.  $t \in I_\rho$ ,

$$u^\perp(t) = \mathcal{T}_{B_\rho} \omega(t)$$

by Theorem C.20. The operator  $\mathcal{T}_{B_\rho}$  is linear, time-independent, and locally Sobolev continuous by Theorem C.21. Hence distributional time differentiation commutes with  $\mathcal{T}_{B_\rho}$ :

$$\partial_t^j u^\perp = \mathcal{T}_{B_\rho} (\partial_t^j \omega)$$

on  $B_r \times I_r$ , for every  $j$  for which the right side is defined. Here the right side is interpreted through the smooth divergence-free Hodge extension of  $\mathcal{T}_{B_\rho}$ . Applying Theorem C.20 to  $\partial_t^j u^\perp$  on each time slice and integrating over  $I_\rho$ , then shrinking to  $I_r$ , gives (11). Since all derivatives of  $\omega$  are smooth and locally square-integrable on compact cylinders, parabolic Sobolev embedding gives  $u^\perp \in C^\infty(B_r \times I_r)$ .  $\square$

**Lemma C.23** (Quotient velocity recovery). *Let  $B_r \times I_r \Subset B_\rho \times I_\rho \Subset Q_R$ , and set*

$$u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u.$$

*Under the hypotheses of Theorem C.5,  $u^\perp$  is smooth in  $B_r \times I_r$ , and for every  $j, k \geq 0$ ,*

$$\|\partial_t^j \nabla^k u^\perp\|_{L^2(B_r \times I_r)} \leq C_{j,k,r,\rho,I_r,I_\rho} \left( \|u\|_{L^2(B_\rho \times I_\rho)} + \sum_{\ell+|\alpha| \leq N_{j,k}} \|\partial_t^\ell \nabla^\alpha f\|_{L^2(B_\rho \times I_\rho)} \right). \quad (12)$$

*If  $\mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on  $B_r \times I_r$ , then  $u$  is smooth there as well.*

*Proof.* Theorem C.15 gives smoothness of  $\omega = \operatorname{curl} u$  on a cylinder between  $B_r \times I_r$  and  $B_\rho \times I_\rho$ , with bounds in terms of  $\|u\|_{L^2}$  and local derivatives of  $f$  on the larger cylinder. Then Theorem C.22 recovers all space and time derivatives of

$$u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u$$

from that local vorticity data. Combining the two estimates gives (12). If the harmonic-kernel component is smooth, then  $u = u^\perp + \mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth. The argument uses only the local  $L^2(B_\rho)$  quotient projection to separate the harmonic-kernel component; no whole-space projection or nonlocal pressure reconstruction is needed.  $\square$

## C.6 Pressure Closure

**Lemma C.24** (Harmonic pressure compatibility). *Let  $B_r \times I_r \Subset B_\sigma \times I_\sigma \Subset B_\rho \times I_\rho \Subset Q_R$ , and use the pressure decomposition of [Theorem C.17](#). Suppose that  $u^\perp$  is smooth on  $B_\sigma \times I_\sigma$  and that the harmonic-kernel component  $\mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth there as well. Then  $\nabla p_{\text{har}}$  is smooth on  $B_r \times I_r$ .*

*Proof.* The two velocity hypotheses imply  $u = u^\perp + \mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on  $B_\sigma \times I_\sigma$ . The Stokes equation gives, in distributions,

$$\nabla p = f - \partial_t u + \Delta u,$$

so  $\nabla p$  is smooth on  $B_\sigma \times I_\sigma$ . By [Theorem C.17](#),  $\nabla p_{\text{CZ}}$  is smooth on  $B_r \times I_r$ . Therefore

$$\nabla p_{\text{har}} = \nabla p - \nabla p_{\text{CZ}}$$

is smooth on  $B_r \times I_r$ . The possible additive time-only representative has zero spatial gradient and does not affect the conclusion.  $\square$

**Lemma C.25** (Pressure gradient closure). *Under the hypotheses of [Theorem C.5](#), assume on  $B_\rho \times I_\rho \Subset Q_R$  that the harmonic-kernel component  $\mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on  $B_r \times I_r \Subset B_\rho \times I_\rho$ . Then, for any pressure representative in the local normalization class,*

$$\nabla p = f - \partial_t u + \Delta u$$

*is smooth in  $B_r \times I_r$ .*

*Proof.* [Theorem C.23](#) gives smoothness of  $u^\perp$ , and the extra hypothesis gives smoothness of the harmonic-kernel component. Hence  $u$  is smooth on  $B_r \times I_r$ , and  $f$  is smooth by hypothesis. The Calderón–Zygmund decomposition of [Theorems C.17](#) and [C.18](#) has already identified the smooth local CZ part and the harmonic pressure component. On every smaller compact subcylinder, [Theorem C.24](#) gives smoothness of the harmonic pressure gradient once both the quotient velocity and the harmonic-kernel component are smooth. Equivalently, the displayed identity is exactly (2) and directly gives the same smooth gradient on  $B_r \times I_r$ . The scalar pressure is determined only up to a function of time, but the gradient is invariant under pressure normalization and locally smooth.  $\square$

**Proposition C.26** (Parasitic mode closure by normalization). *Let  $B_r \times I_r \Subset B_\rho \times I_\rho \Subset Q_R$ , and let  $(u, p, f)$  satisfy the hypotheses of [Theorem C.5](#). Set*

$$h = \mathbf{P}_{\mathcal{H}(B_\rho)}u.$$

*Then the harmonic-kernel component  $h$  is handled as follows:*

1. *The normalized representative*

$$(u^\perp, p^\perp)$$

*constructed in [Theorem C.3](#) satisfies  $\mathbf{P}_{\mathcal{H}(B_\rho)}u^\perp = 0$ , and therefore  $u^\perp$  and  $\nabla p^\perp$  are smooth on  $B_r \times I_r$ .*

2. *If one also wants regularity of the original raw representative, then the extra compatibility condition is exactly that  $h$  vanish or be smooth on  $B_r \times I_r$ . Under that condition,*

$$u \in C^\infty(B_r \times I_r), \quad \nabla p \in C^\infty(B_r \times I_r).$$

3. If  $h$  is not normalized away and no smoothness input for  $h$  is available, full regularity of that raw representative is not a valid conclusion from the local Stokes hypotheses alone; the valid conclusion is regularity of the normalized representative.

*Proof.* The first item follows from [Theorem C.3](#): the normalized representative solves the same Stokes system and has zero projected harmonic-kernel component. [Theorem C.23](#) gives smoothness of  $u^\perp$ , and [Theorem C.25](#) gives smoothness of  $\nabla p^\perp$  because the projected kernel of the normalized representative is zero. For the second item,

$$u = u^\perp + h$$

is smooth on  $B_r \times I_r$ . Since

$$p^\perp = p + \partial_t \Phi, \quad \nabla \Phi = h,$$

we have

$$\nabla p = \nabla p^\perp - \partial_t h,$$

which is smooth if  $h$  is smooth. The third item is the counterexample in [Theorem C.4](#): a nonsmooth time-dependent harmonic-gradient velocity has zero vorticity and is absorbed by the pressure. Thus local normalization is the step that closes the argument for the normalized representative, while smoothness of  $h$  is the extra condition needed for the original raw representative.  $\square$

**Proposition C.27** (Regularity of the normalized representative). *Let  $(u, p, f)$  solve (2) on  $Q_R$  with  $u \in L^2_{\text{loc}}(Q_R)$  and  $f \in C^\infty(Q_R)$ , and suppose the representative under consideration is harmonic-kernel normalized on each nested intermediate cylinder. Then  $u$  is smooth on every compactly contained subcylinder  $Q' \Subset Q_R$ .*

*Proof.* Every candidate singular point of the normalized representative leads to the vorticity argument of [Theorem C.12](#) and the ordered alternatives of [Theorem C.11](#). By [Theorem C.15](#),  $\omega = \text{curl } u$  is smooth on smaller cylinders. [Theorem C.23](#) then gives smoothness of the quotient representative  $u^\perp$ . Since the harmonic-kernel component has been normalized away,  $u^\perp = u$  and the solution is smooth near every candidate singular point.  $\square$

**Proposition C.28** (Absence of interior singular points). *Let  $(u, p, f)$  satisfy the hypotheses of [Theorem C.5](#). Then  $u$  has no candidate singular points for the locally normalized representative. The same holds for the raw representative when its harmonic-kernel component is zero or smooth on the relevant nested cylinder.*

*Proof.* Suppose  $z_* \in Q_{R/4}$  is a candidate singular point for the normalized representative. By [Theorem C.12](#), either the normalized solution is already regular near  $z_*$ , or  $\omega = \text{curl } u$  satisfies the local heat equation. In the latter case, [Theorem C.15](#) smooths the vorticity, and [Theorems C.17](#) and [C.18](#) decompose the pressure into a smooth Calderón–Zygmund part and a harmonic part. [Theorem C.23](#) recovers smoothness of  $u^\perp$  near  $z_*$ . [Theorem C.26](#) then gives smoothness of the normalized representative and its pressure gradient near  $z_*$ , contradicting the assumption. When the harmonic-kernel component is zero or smooth, the same proposition transfers the conclusion to the raw representative.  $\square$

*Proof of [Theorem C.5](#).* Fix  $B_r \times I_r \Subset B_\rho \times I_\rho \Subset Q_R$ . The estimate (3) is [Theorem C.23](#), whose proof combines the local heat smoothing of  $\omega = \text{curl } u$  with the time-parametrized quotient div-curl map. Thus  $u^\perp = (I - \mathbf{P}_{\mathcal{H}(B_\rho)})u$  is smooth on  $B_r \times I_r$ , and the estimate uses only data on the selected larger cylinder.

If the harmonic-kernel component  $h = \mathbf{P}_{\mathcal{H}(B_\rho)}u$  is smooth on  $B_r \times I_r$ , then  $u = u^\perp + h$  is smooth there. The Stokes equation then gives

$$\nabla p = f - \partial_t u + \Delta u$$

as a smooth vector field on the same smaller cylinder. In the normalized case  $h = 0$ , this gives full smoothness of the normalized representative and its pressure gradient directly.  $\square$

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